

Reviewed Preprint

v1 • July 14, 2025

Not revised

Reviewed Preprint

v2 • February 18, 2026

Revised by authors

✉ For correspondence:

hiromut@m.tohoku.ac.jp

Competing interests:

No competing interests declared

Funding:

See [page 21](#)

Reviewing editor:

P Robin Hiesinger, Institute for Biology Free University Berlin, Germany

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Profiling presynaptic scaffolds using split-GFP reconstitution reveals cell-type-specific spatial configurations in the fly brain

Hongyang Wu¹, Yoh Maekawa², Sayaka Eno¹, Shu Kondo³, Nobuhiro Yamagata^{1,4}, Hiromu Tanimoto^{1,2} ✉

¹Graduate School of Life Sciences, Tohoku University, Sendai, Japan • ²Department of Biology, Faculty of Science, Tohoku University, Sendai, Japan • ³Department of Biological Science and Technology, Tokyo University of Science, Tokyo, Japan • ⁴Department of Life Science, Graduate School of Engineering Science, Akita, Akita University, Akita, Japan

eLife Assessment

This **important** work introduces a splitGFP-based labeling tool with an analysis pipeline for the synaptic scaffold protein bruchpilot, with tests in the adult *Drosophila* mushroom bodies, a learning center in the *Drosophila* brain. The evidence supporting the conclusions is **convincing**.

<https://doi.org/10.7554/eLife.107663.2.sa4>

Abstract

Characterization of intracellular synapse heterogeneity aides to understand the intricate computational logic of neuronal circuits. Despite recent advances in connectomics, the spatial patterns of synapses and their inter-individual variability remain largely unknown. Using directed split-GFP reconstitution, we achieved visualization of endogenous Bruchpilot (Brp), a presynaptic active zone (AZ) scaffold protein, in a cell-type-specific manner. By developing a high-throughput quantification pipeline, we profiled AZ structures in identified neurons of the mushroom body circuit, where intracellular synaptic patterns are crucial due to compartmentalized connectivity. Quantitative characterization of the pattern of Brp clusters across multiple individuals revealed cell-type-dependent synapse heterogeneity and stereotypy. Furthermore, we discovered previously unidentified sub-compartmental synapse configuration and its transient structural plasticity triggered by associative learning. These profiles thus uncovered multi-layered spatial configurations of AZs, from stereotyped overall AZ distribution patterns, to local arrangements of neighboring synapses.

Introduction

The brain is a complex network of neurons, and understanding their wiring strategies provides crucial insights into their roles in neuronal computation. Release sites in the presynaptic terminal of *Drosophila* neurons are decorated with electron-dense projections composed of active-zone (AZ) scaffolds that are essential for efficient synaptic transmission (Akbergenova et al., 2018 [DOI](#); Fouquet et al., 2009 [DOI](#); Kittel et al., 2006 [DOI](#); Matkovic et al., 2013 [DOI](#); Newman et al., 2022 [DOI](#); Paul et al., 2015 [DOI](#); Wagh et al., 2006 [DOI](#)). These dense projections serve as a synapse marker enabling the whole-brain connectomes, comprehensive mapping of synaptic connections at the electron microscopic level (Dorkenwald et al., 2024 [DOI](#); Scheffer et al., 2020 [DOI](#); Schlegel et al., 2024 [DOI](#); Takemura et al., 2024 [DOI](#); Zheng et al., 2018 [DOI](#)). While connectome-based approaches catalyzed the exploration of cell type distinctions, circuit motifs, and neurotransmitters, their application to comparisons across different conditions is still challenging due to the limited throughput of data

acquisition and dense reconstruction. Since the synaptic structure and connectivity reflect substantial variability of development and experiences (Fernández et al., 2008; Gilestro et al., 2009; Kremer et al., 2010; Sachse et al., 2007; Schlegel et al., 2024; Turrel et al., 2022; Zhang et al., 2018), cross-individual comparisons are necessary. Fluorescence labeling offers high throughputs and contents, especially when comparing synaptic organizations across cell-types (Mosca and Luo, 2014) or brain regions (Gao et al., 2019). Nevertheless, characterizing endogenous synaptic proteins in specific cells using fluorescence-labeling remains challenging in the central nervous system (CNS). This has driven growing interest in designing cell-type-specific fluorescence-tagging strategies for endogenous synaptic protein (Chen et al., 2014).

The circuit of *Drosophila* mushroom bodies (MBs) plays a central role in olfactory associative learning (Davis, 2023). The major intrinsic MB neuron, known as Kenyon cells (KCs) respond to odor stimulation and synapse onto the spatially segregated dendrites of MB output neurons (MBONs) that divide the MB into compartments. Distinct types of dopaminergic neurons (DANs) synapse onto specific compartments and thereby modulate KC-MBON synapses in the corresponding compartments (Aso et al., 2014). Presynaptic calcium levels of these DANs undergo sub-compartmental GABAergic modulation and inform memory specificity, postulating the distinct synapse structures at individual release sites (Yamagata et al., 2021). Two types of giant interneurons, the anterior paired lateral neuron (APL) and the dorsal paired medial (DPM) neuron extensively ramify across the MB lobes and provide recurrent modulations to KC synapses (Haynes et al., 2015; Keene et al., 2006, 2004; Lin et al., 2014; Liu and Davis, 2009; Pitman et al., 2011; Waddell et al., 2000; Yu et al., 2005). Considering their comprehensive projections throughout the MB lobes, the modulation of local circuits requires intracellular synaptic tuning. These studies together underscore the importance to spatially distinguish individual synapses within a cell. However, the high AZ density in the MB lobes precluded conventional approaches of fluorescent microscopy from profiling synaptic structures of specific neurons (Scheffer et al., 2020).

Here, we present a spatial analysis of individual AZs within the MB circuit at the single-cell resolution, achieved by cell-type-specific visualization of the endogenous active zone protein Bruchpilot (Brp) using the CRISPR/Cas9-mediated split-GFP tagging system. *Drosophila* ELKS/CAST/ERC family member Brp plays a central role in molecular assemblies at AZs by accumulating calcium channels and synaptic vesicles (Fouquet et al., 2009; Hallermann et al., 2010; Kittel et al., 2006; Matkovic et al., 2013; Wagh et al., 2006). Therefore, Brp enrichment serve as a structural proxy suited for estimating synapse function, such as release probability at single AZs (Akbergenova et al., 2018; Newman et al., 2022). We developed a high-throughput quantification pipeline to systematically profile Brp clusters of individual AZs in different MB-innervating neurons. Characterizing the distinct parameters and localization of Brp clusters revealed AZ distribution stereotypy across individuals and significant synaptic heterogeneity within single neurons. These cell-type-specific synapse profiles suggest that AZs are spatially organized at multiple scales, ranging from interindividual stereotypy to neighboring synapses in the same cells.

Results

Establishing an experimental system for cell-type-specific profiling of active zone structures

As the molecular assembly of synapses is reported to be sensitive to the dosage of Brp, we visualized endogenous Brp instead of transgenic expression of tagged Brp (Huang et al., 2020). To label endogenous Brp specifically in designated cell types, we employed the split-GFP tagging system (Kamiyama et al., 2016; Kondo et al., 2020). Using CRISPR/Cas9, we inserted the GFP₁₁ fragment (the eleventh β -strand of the super-folder GFP) just prior to the stop codon of *brp* gene. The GFP₁₋₁₀ fragment is expressed in cells of interest by the GAL4/UAS system. The self-assembly of the split-GFP fragments is therefore directed only in GAL4-expressing cells and visualizes endogenous Brp through reconstituted GFP fluorescence (Figure 1A).

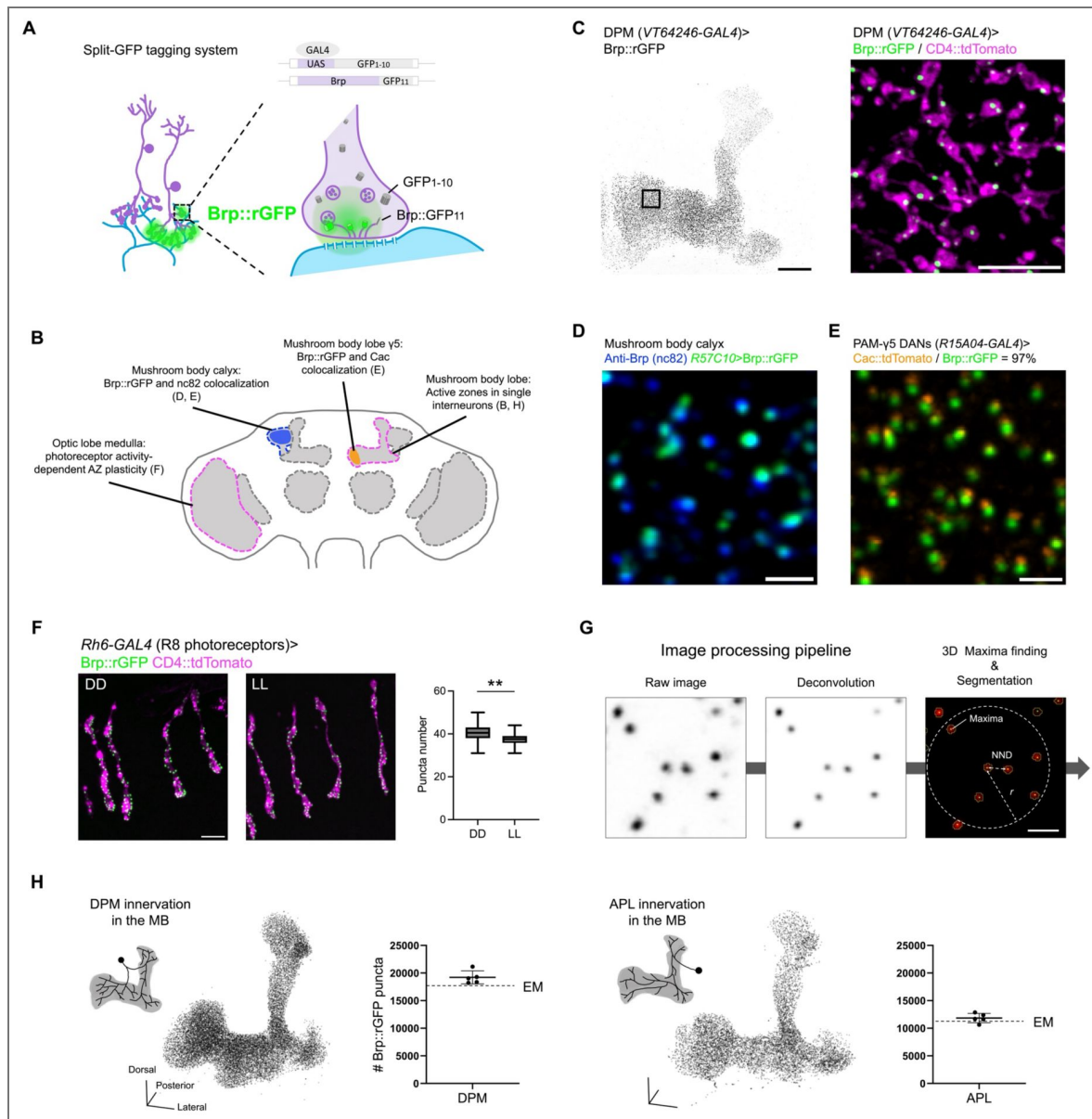


Figure 1. Spatial characterization of Brp::rGFP clusters at single-AZ level.

(A) Schematic of the split-GFP tagging strategy. The GFP₁₁ fragment is inserted by CRISPR/Cas9 just prior to the stop codon of *brp*, while GFP₁₋₁₀ is expressed in specific cell types via GAL4-UAS system. (B) Brp::rGFP signals in DPM. The entire MB lobe structure is shown. The black box in the left panel indicates the zoom-in area (right panel). GFP₁₋₁₀ and CD4::tdTomato (magenta) were co-expressed using VT64246-GAL4. Scale bar, 20 μ m (left); 5 μ m (right). (C) Schematic showing brain regions and cell types analyzed in Figure 1C-F and H. (D) Brp::rGFP (green) and anti-Brp immunostaining signals (blue) in the mushroom body calyx. GFP₁₋₁₀ was expressed using pan-neuronal driver R57C10-GAL4. Scale bar, 1 μ m. (E) Brp::rGFP (green) and Cac::tdTomato (orange) in PAM-y5 dopamine neuron terminals. GFP₁₋₁₀ and Cac::tdTomato were co-expressed using R15A04-GAL4. Scale bar, 1 μ m. (F) Brp::rGFP (green) in R8 photoreceptors upon constant dark or light conditions. GFP₁₋₁₀ and CD4::TdTomato (magenta) were co-expressed using Rh6-GAL4. Brp::rGFP cluster number of each R8 photoreceptor terminal in the medulla were quantified. $n = 30$ (3 brain samples, 10 terminals from each brain). Welch's t test. Scale bar, 5 μ m. (G) Image processing pipeline. Raw images were processed with image deconvolution to improve the signal quality. 3D maxima (green pixels) are detected for each Brp::rGFP cluster. 3D ROIs are indicated by yellow circles. Nearest neighbor distance (NND), r is defined as $3 \times \text{mean NND}$. Scale bar, 1 μ m. (H) Number of Brp::rGFP clusters detected in DPM and APL. The dashed lines indicate the number of electron dense projections identified in the hemibrain connectome for DPM (17530) and APL (10789) on the MB lobes and peduncle (Scheffer et al., 2020). Schematics illustrate the innervation patterns of DPM and APL neurons. For both DPM and APL, $n = 5$. Lower panels show the representative 3D reconstructions of Brp::rGFP clusters in DPM and APL. Scale bars, 20 μ m. Data were presented as mean $\pm$ s.e.m.

As a proof of principle, we first validated the localization and plasticity of split-GFP tagged Brp (Figure 1B [↗](#)). To verify tagged Brp localization in an individual neuron, we directed GFP reconstitution and expressed a plasma membrane marker in the serotonergic DPM neuron using *VT64246-GAL4* (Figure 1C [↗](#)). Confocal microscopy revealed Brp::reconstituted GFP (Brp::rGFP) signals only to the MB lobes, aligning with DPM terminals (Waddell et al., 2000 [↗](#)). These signals were distributed as discrete clusters within plasma membrane varicosities resembling presynaptic boutons (Figure 1C [↗](#)), suggesting that Brp::rGFP clusters labeled individual AZs in the DPM neuron. AZ localization of Brp::rGFP was further validated by co-localization with anti-Brp immunostaining in the calyx and the voltage-gated calcium channel subunit Cacophony (Cac) in the PAM dopaminergic neurons (Figure 1D-E [↗](#)) (Kawasaki et al., 2004 [↗](#); Kittel et al., 2006 [↗](#)). We found that GFP₁₁ tagging did not affect *brp* mRNA levels, but slightly decreased protein expression without altering overall expression pattern in the brain (Figure 1 – figure supplement 1 [↗](#) [↗](#)). Brp::rGFP successfully visualized previously reported light-induced AZ plasticity in the photoreceptors (Sugie et al., 2015 [↗](#)). Constant light exposure reduced the numbers of Brp::rGFP clusters in the R8 terminals in the medulla, compared to constant darkness (DD) (Figure 1F [↗](#)). This was further substantiated by increased Brp::rGFP cluster intensities upon knocking-down *rab3* expression (Figure 1 – figure supplement 3 [↗](#)), recapitulating Rab3-dependent regulation of Brp allocation in the presynaptic terminals of the motor neurons (Graf et al., 2009 [↗](#)).

To systematically profile Brp::rGFP clusters, we developed an image processing pipeline (Figure 1A [↗](#)). In brief, image deconvolution was applied to reduce out-of-focus light and enhance signal sharpness (Figure 1 – figure supplement 4 [↗](#)). Individual Brp::rGFP clusters were then segmented using the 3D suite plugin in Fiji (Ollion et al., 2013 [↗](#)), generating the 3D region-of-interests (3D ROIs) that encircle each cluster. Brp::rGFP signal intensity, volume and location of individual clusters were measured using the 3D ROIs. To optimize the segmentation, we systematically varied the parameters and calculated the F-score to measure detection accuracy against manually defined ground truth (see Materials and Methods). The F-score is near 1.0 in segmenting Brp::rGFP clusters within DPM and APL neurons and achieved > 0.90 in KCs with certain combinations of segmentation parameters (Figure 1 – figure supplement 5 [↗](#)).

To confirm that Brp clusters represent AZs, we quantified the numbers of Brp::rGFP clusters in DPM and APL neurons specifically and compared them to the connectome data. We found that cluster counts closely matched the number of AZs annotated by the hemibrain connectome (Figure 1C [↗](#)) (Scheffer et al., 2020 [↗](#)). This result validated our method that accurately isolated Brp::rGFP clusters corresponding to single AZs.

AZ profiles within KCs are compartmentally heterogeneous

Brp accumulation at individual AZs is known to be heterogeneous even within a single motor neuron (Akbergenova et al., 2018 [↗](#); Ehmann et al., 2014 [↗](#); Gratz et al., 2019 [↗](#); Paul et al., 2015 [↗](#)). To study such AZ heterogeneity in CNS neurons, we quantified the signal intensity of individual Brp::rGFP clusters in MB lobes. Using KC subtype-specific drivers to direct split-GFP reconstitution in γ KC (*MB009B-GAL4*), α'/β' KCs (*MB370B-GAL4*) and α/β KCs (*MB008B-GAL4*), we found that within-cluster Brp::rGFP intensities are highly diverse in all KC subtypes (Figure 2A [↗](#)).

To visualize spatial patterns of Brp::rGFP intensity in each KC subtype, we reconstructed 3D distribution of Brp::rGFP clusters and color-coded them by signal intensities (Figure 2B [↗](#)). We found that clusters showed noticeable intensity differences between compartments in all subtypes. Even adjacent compartments such as γ_1 and γ_2 , β'_2 and β'_1 , α_2 and α_3 , showed drastic differences (Figure 2B [↗](#)). Importantly, these compartmental patterns were stable across individuals (Figure 2C [↗](#)) (Wu et al., 2025 [↗](#)). This result suggests that AZs in different compartments have distinct structures. Since individual KCs of each subtype synapse onto all compartments (Scheffer et al., 2020 [↗](#); Schlegel et al., 2023 [↗](#); Takemura et al., 2024 [↗](#); Zheng et al., 2018 [↗](#)), this Brp compartmental heterogeneity is likely formed intracellularly.

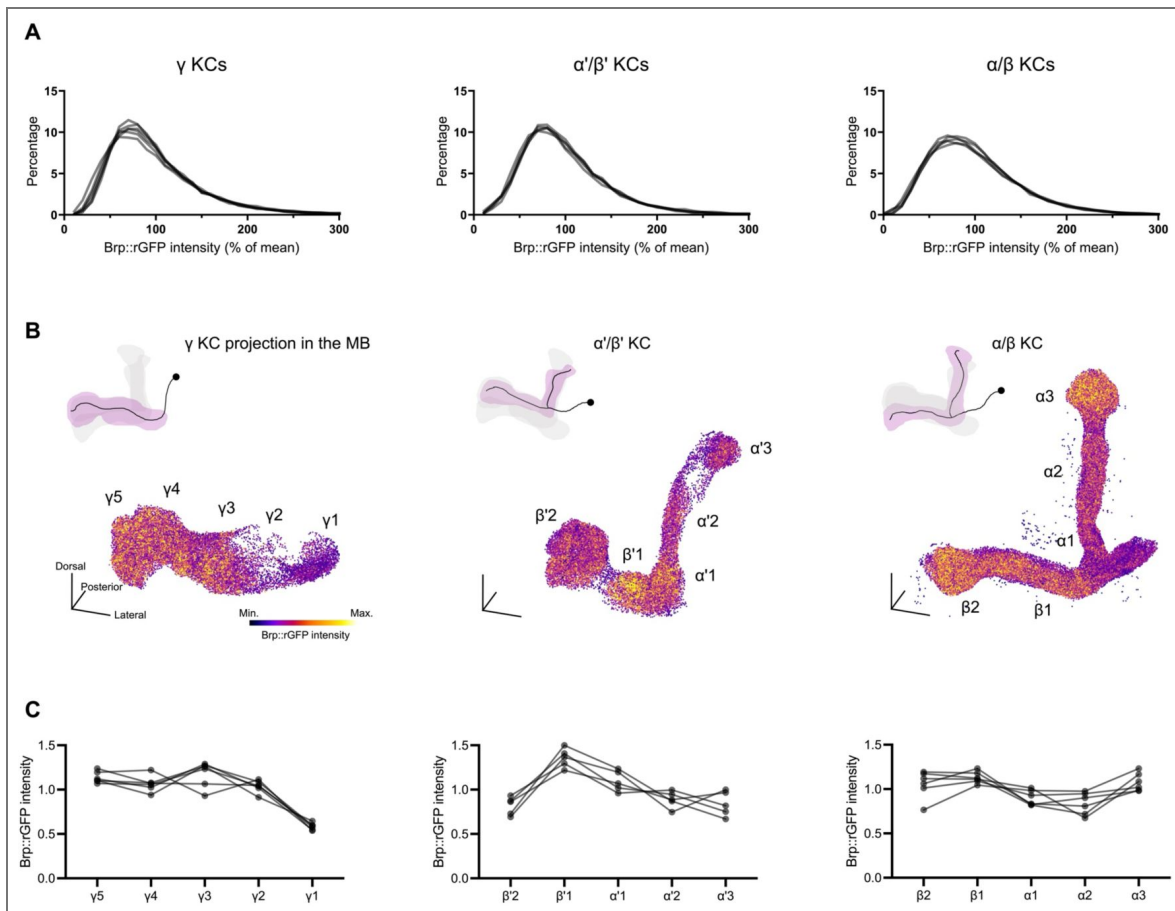


Figure 2. Compartmentalized AZ structures of KCs.

(A) Histograms of the signal intensity of individual Brp::rGFP clusters from three KC subtypes. Values are normalized by the mean intensity in each data set (each sample). Each line represents the histogram of one independent sample. γ KCs (*MB009B-GAL4*), $n = 6$; α/β KCs (*MB370B-GAL4*), $n = 5$; α'/β' KCs (*MB008B-GAL4*), $n = 5$. (B) 3D reconstructions of Brp::rGFP clusters in three KC subtypes, colored by the Brp::rGFP intensity. The approximate locations of compartments are as indicated. Schematics illustrate the innervation patterns of different KC subtypes. Min. and Max. were set to represent the lowest and highest 5% of Brp::rGFP intensity value in the dataset respectively. Scale bars, 20 μ m. (C) Signal intensity of Brp::rGFP clusters in each compartment. Brp::rGFP clusters were quantified compartmentally. Medians in different compartments are showed as the ratio against the average of five compartments in the corresponding KC subtype. Each line represents one independent sample.

Intracellular heterogeneity of Brp concentration at individual AZs and cell-type dependent diversity

Super-resolution microscopy has shown that the Brp molecular density at individual AZs is dynamically adjusted in motor neurons (Ghelani et al., 2023 [↗](#); Mrestani et al., 2021 [↗](#)). We hypothesized that the amount of Brp does not necessarily correlate with AZ cluster size. To verify this, we calculated the Brp concentration by quantifying both the volume (number of voxels within a 3D ROI) and the Brp::rGFP intensity of each cluster. We found that clusters with similar sizes can exhibit vastly different signal intensities in the DPM neuron, suggesting the heterogeneity of Brp concentration at individual AZs (Figure 3A [↗](#)).

Correlation analysis between the Brp::rGFP intensity and cluster volume revealed a substantial variability among cell types (Figure 3B-E [↗](#)). There were even significant differences among KC subtypes. These results suggest cell-type dependent variability of the Brp concentration of individual clusters.

Stereotypy of intracellular distribution of AZs

To study the spatial distribution of Brp clusters within single neurons, we focused on DANs. *MB504B-GAL4* labels four types of single PPL1 DANs projecting to distinct compartments (Vogt et al., 2014 [↗](#)). Using this driver, we analyzed Brp::rGFP distribution in each compartment and found it uneven in terminals of both PPL1- $\alpha 2$ and $\alpha 3$ (Figure 4A-B [↗](#)). To better characterize cluster distribution, we calculated the “AZ density” defined as the number of Brp::rGFP clusters surrounding a particular cluster within a specified radius (see Figure 1A [↗](#) and Materials and Methods for detail). By color-coding Brp::rGFP clusters according to their AZ density, we found that AZs in PPL1- $\alpha 3$ were more localized to the core of the $\alpha 3$ compartment. This pattern was consistent across individuals (Figure 4C-D [↗](#)), suggesting a stereotyped AZ distribution within PPL1- $\alpha 3$ neurons.

We extended the analysis to two other single pairs of neurons innervating the MBs, the DPM and APL neurons. Visualizing AZ density revealed a stereotyped and compartment specific distribution in the DPM neuron (Figure 5A [↗](#)). Specifically, the AZ density was constantly high in the α'/β' lobes across all individuals, supporting the known functional importance of DPM branches in the α'/β' lobes (Keene et al., 2006 [↗](#)). In contrast, this compartmental pattern was less pronounced in the APL neuron. Notably, the ratio of AZ density between $\alpha 3$ and $\alpha'3$ is strikingly different among individuals (Figure 5B [↗](#)). Taken together, these data suggest that the individual variability in AZ distribution depends on the cell type.

Regulation of spatial configuration of AZs

Characteristics of Brp clusters and AZ distribution in KCs revealed spatial organization at the level of compartments. Recent studies further showed intra-compartmental variability of synaptic plasticity upon associative learning, prompting us to analyze sub-compartmental organization of AZs (Bilz et al., 2020 [↗](#); Davidson et al., 2023 [↗](#)). We thus examined potential local patterns among AZs in proximity by calculating the correlation between nearest neighbor clusters in terms of their Brp::rGFP intensities (Figure 6A-B [↗](#)). This analysis revealed strong correlations in KCs and APL but not in DPM (Figure 6C [↗](#)).

We further analyzed the sub-compartmental AZ patterns by examining the spatial distribution of Brp::rGFP intensity. To reduce the high frequency variability of Brp::rGFP intensities (Figure 2B [↗](#)), we calculated the “local intensity” for each cluster by applying mean filtering (the average Brp::rGFP intensity of all surrounding clusters within a specified radius r , see Materials and Methods for detail). By visualizing the local intensity in γ KCs, we identified AZ “hot spots”, sub-compartment-sized small groups of AZs with high local intensities, on top of the compartmental differences (Figure 6D [↗](#)). Since Brp molecules are clustered more in AZ-dense boutons of a single motor neuron (Paul et al., 2015 [↗](#)), we hypothesized that these hot spots correspond to regions with higher AZ densities. Indeed, we found that hot spots appear in high AZ densities regions in γ KCs (Figure 6D [↗](#)). Consistently, correlation between the local intensity and the AZ density was

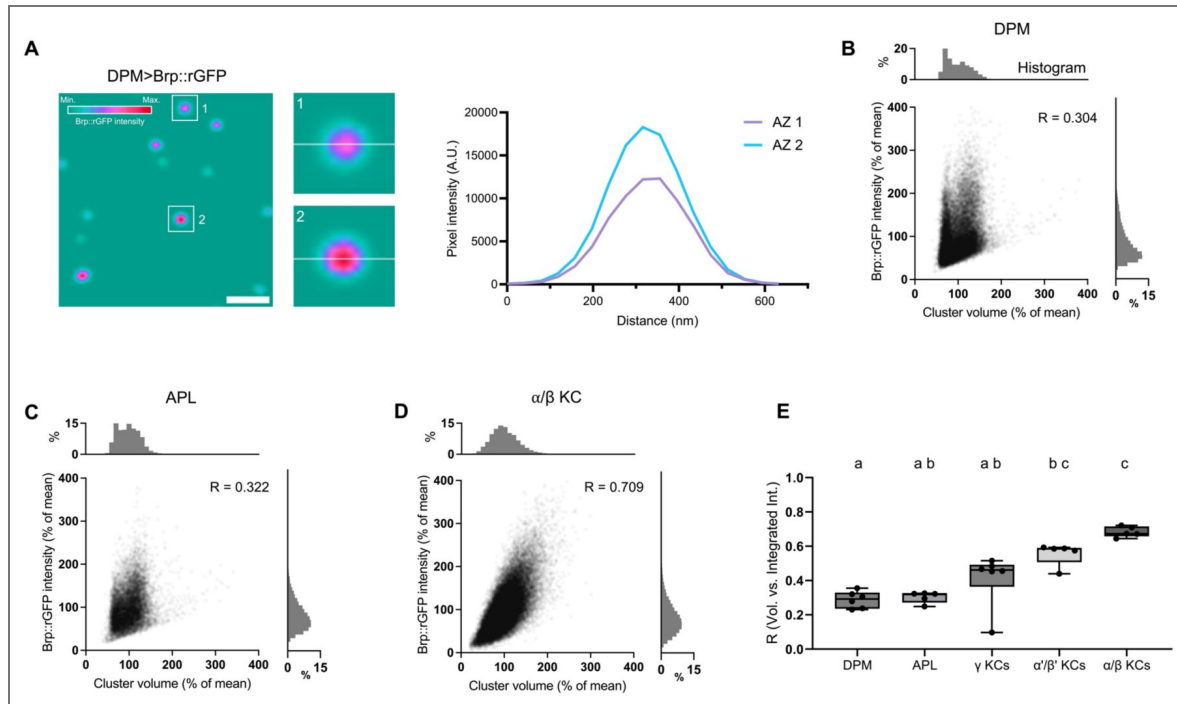


Figure 3. The variability of Brp::rGFP concentrations depends on cell type.

(A) Brp::rGFP clusters with distinct intensities but similar size in DPM. The left panel shows a cropped image of Brp::rGFP in DPM. White boxes indicate areas zoomed-in in panels 1 and 2. Transparent white lines in panels 1 and 2 show the lines on which the intensity profiles were plotted. Intensity profiles were plotted for AZ1 (magenta) and AZ2 (blue) respectively. (B-D) Scatter plots showing the correlation between the Brp::rGFP intensity and cluster volume in DPM, APL and α/β KCs. Data of three representative samples were shown. Pearson correlation coefficient R were calculated for each sample. (E) Correlations between the intensity and volume of Brp::rGFP clusters in different cell types. DPM ($n = 6$), APL ($n = 5$), γ KCs ($n = 6$, *MB009B-GAL4*), α'/β' KCs ($n = 5$, *MB370B-GAL4*), α/β KCs ($n = 5$, *MB008B-GAL4*). DPM vs. α'/β' KCs: $P = 0.0253$; DPM vs. α/β KCs: $P = 0.0009$; APL vs. α/β KCs: $P = 0.0035$; γ KCs vs. α/β KCs: $P = 0.0418$. Values marked with different lowercase letters represent significant difference ($P < 0.05$); Data were presented as box plots showing center (median), whiskers (Min. to Max.).

Figure 4. Stereotyped AZ distribution of PPL1- $\alpha 3$ DAN.

(A) Brp::rGFP in PPL1 DANs. UAS-GFP₁₋₁₀ was expressed using *MB504B-GAL4*. Dashed line marks the rough boundary between $\alpha 3$ and $\alpha 2\alpha'2$ compartments. Schematic illustrates the innervation patterns of PPL1 DANs within the MB. Scale bar, 20 μm . (B) Brp::rGFP intensity profiles of $\alpha 3$ and $\alpha 2\alpha'2$ compartments. Left panels show the max-projection images of $\alpha 3$ and $\alpha 2\alpha'2$ optical coronal sections. Transparent gray stripes indicate areas where intensity profiles are plotted in the right panels. Scale bars, 10 μm . (C) 3D reconstruction colored by the AZ density in PPL1 DANs. Color scale: Min. = 0, Max. = 40. Dashed line indicates the rough boundary between $\alpha 3$ and $\alpha 2\alpha'2$ compartments. Scale bar, 20 μm . (D) Stereotyped AZ distribution pattern in PPL1- $\alpha 3$ across individuals. 3D reconstructions show the AZ density across different brain samples. Scale bar, 20 μm .

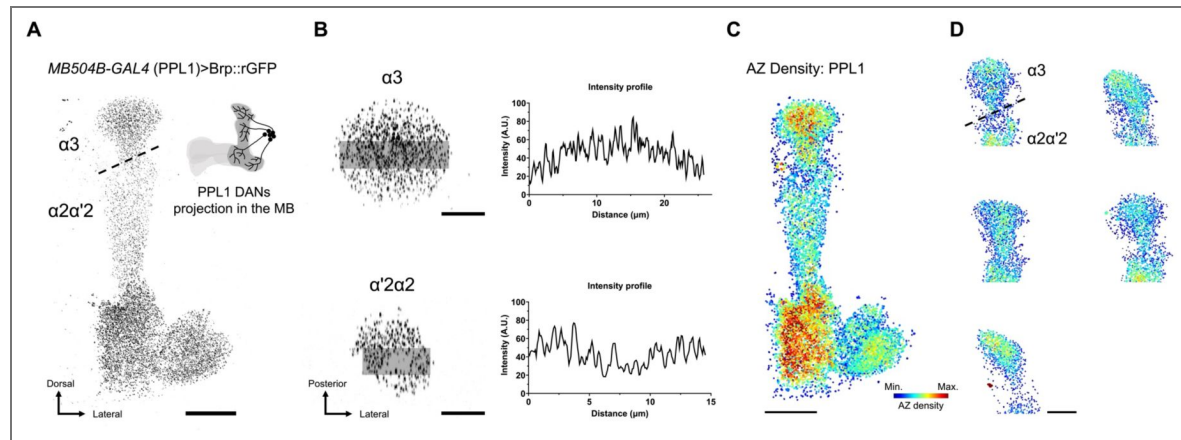
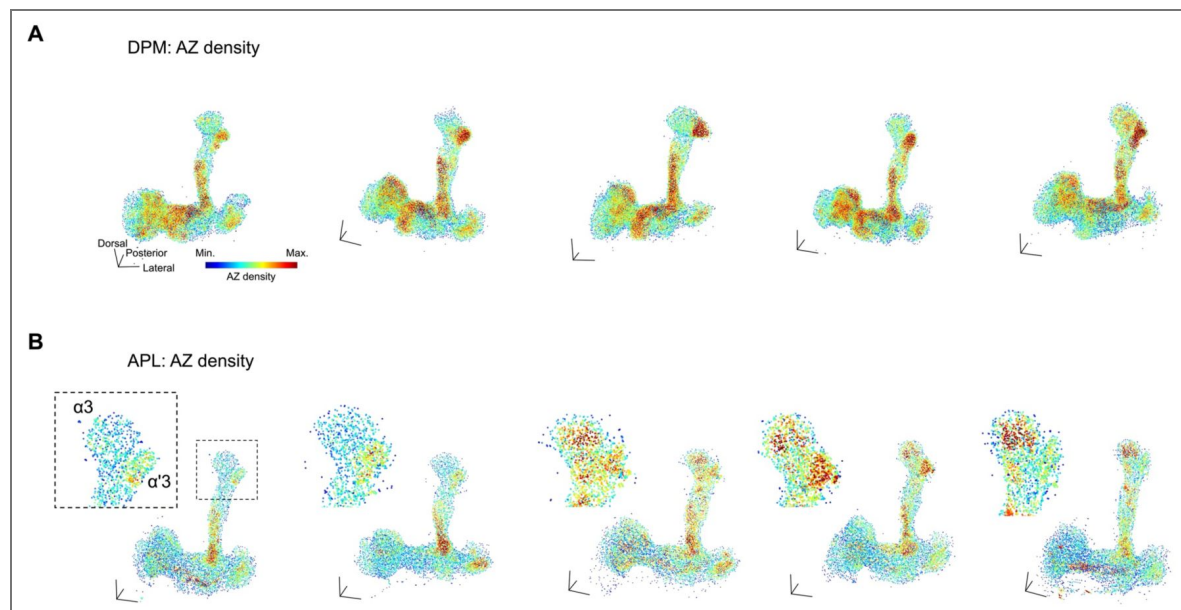


Figure 5. Cell-type-specific stereotypy of AZ spatial distributions.

(A) 3D reconstructions of Brp::rGFP clusters in DPMs, colored by AZ density. Color scale: Min. = 0, Max. = 35. Black arrows indicate consistently high AZ density regions across brain samples. Scale bars, 20 μm . (B) 3D reconstructions of Brp::rGFP clusters in APLs, colored by AZ density. Color scale: Min. = 0, Max. = 30. Black dashed square indicates the area zoomed in. APL reconstructions are arranged from left to right according to the overall AZ density in $\alpha 3$.



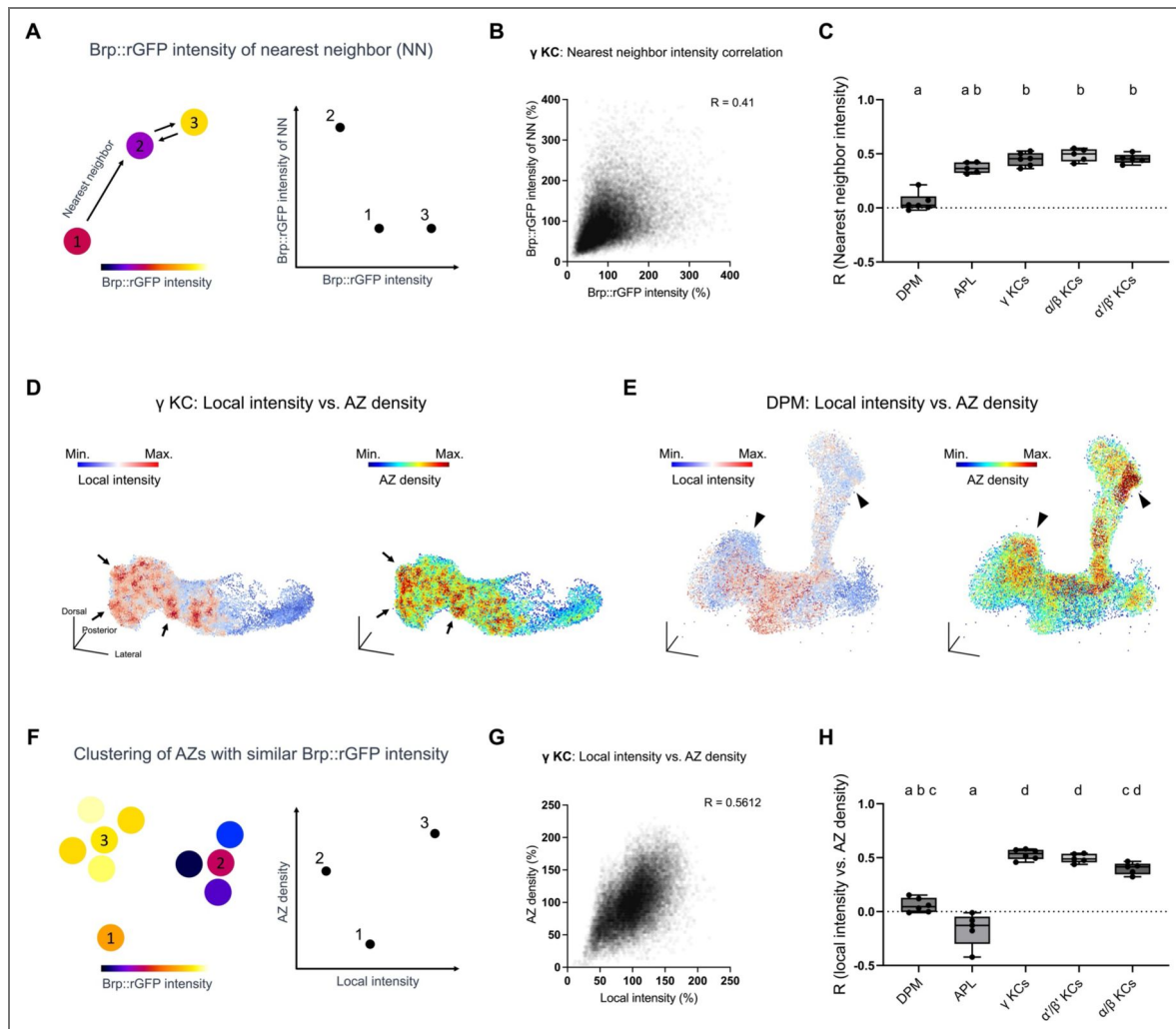


Figure 6. Local intensity analysis revealed sub-compartmental AZ structures.

(A) Brp::rGFP correlation analysis between nearest neighbors (NN). The intensity of a cluster is plotted on the x-axis and the intensity of its nearest neighbor is plotted on the y-axis. High correlation indicates nearest neighbors have similar Brp::rGFP intensity. (B) Scatter plots showing the Brp::rGFP intensity correlation between nearest neighbors in a representative sample of γ KCs. Values were normalized to the mean. Pearson correlation coefficient R is shown. (C) Correlation of Brp::rGFP intensities between nearest neighbor AZs in different cell types. Brp::rGFP intensities were log transformed. DPM vs. γ KCs: $P = 0.0066$; DPM vs. α'/β' KCs: $P = 0.0027$; DPM vs. α/β KCs: $P = 0.0066$; Scale bar: 1 μ m. (D) 3D reconstructions of Brp::rGFP clusters in a γ KC sample, colored by local intensity and AZ density. Black arrows indicate areas with both high local intensities and high AZ densities. Min. and Max. were set to represent the lowest and highest 5% in local intensity value respectively. Color scale of AZ density: Min. = 0 and Max. = 35. (E) 3D reconstructions of Brp::rGFP clusters in a DPM neuron, colored by local intensity and AZ density. Black triangle arrows indicate areas with high AZ densities but low local intensities. Min. and Max. are set to represent the lowest and highest 5% in local intensity value respectively. Color scale of AZ density: Min. = 0, Max. = 35. (F) Correlation analysis between local intensity and AZ density. The local intensity of a cluster is plotted on the x-axis and its AZ density is plotted on the y-axis. (G) Scatter plots showing the correlation between Brp::rGFP local intensity and AZ density in a representative sample of γ KCs. Values were normalized to the mean. Pearson correlation coefficient R is shown. (H) Correlations between AZ density and local intensity in different cell types. DPM ($n = 6$), APL ($n = 5$), γ KCs ($n = 6$, *MB009B-GAL4*), α'/β' KCs ($n = 5$, *MB370B-GAL4*), α/β KCs ($n = 5$, *MB008B-GAL4*). DPM vs. γ KCs: $P = 0.0171$; DPM vs. α'/β' KCs: $P = 0.0171$; APL vs. γ KCs: $P = 0.0014$; APL vs. α'/β' KCs: $P = 0.0014$; APL vs. α/β KCs: $P = 0.0171$.

high in all KC subtypes. In contrast, for DPM and APL neurons, the local intensity of Brp::rGFP was not associated with AZ density (Figure 6E-H). Taken together, these results suggest previously unidentified sub-compartmental synaptic configuration organized across individual KCs.

Associative learning re-organizes sub-compartmental synaptic configuration

Sub-compartmental synaptic configuration may undergo experience-dependent changes, such as through associative learning (Turrel et al., 2022; Zhang et al., 2018). To examine such structural plasticity in local AZ configurations (Figure 6D, F-H), we visualized Brp::rGFP specifically in KCs using *R13F02-GAL4* (Figure 7A). This induction did not alter MB morphology and short-term memory compared to wild-type (Figure 7 – figure supplement 1). We presented flies with odor (4-methylcyclohexanol) with a concomitant (paired) or shifted (unpaired) electric shock (Fig. 7B). Since both paired and unpaired groups were exposed to the same odor and electric shock, the difference between these groups purely represents the effect of association. As single odor conditioning induced memory decaying gradually over one day (Figure 7B), we measured the correlation between the AZ density and local intensity of individual Brp::rGFP clusters (Figure 6F-H) at different time points after conditioning (from 3 min to 1 day). Strikingly, we found that associative learning induced a transient modification to the KC synaptic configurations in a compartment-specific manner (Figure 7C). This learning-induced structural plasticity was specific at 90 min after conditioning and disappeared within one day (Figure 7D), consistent with previous studies reporting short-lived AZ remodeling in the MBs by conditioning (Turrel et al., 2022; Zhang et al., 2018). These results showcase the advantage of split-GFP system in reporting the dynamics of local synapse subsets with high throughputs and contents. Altogether, we suggest that associative learning induces transient memory traces of local structural plasticity in AZ configuration.

Discussion

By leveraging the CRISPR/Cas9 genome editing of the *brp* locus and split-GFP technique, we systematically profiled presynaptic structures of MB neurons in a cell-type-specific manner. This approach enabled high-throughput analysis of multiple brain samples and detailed characterizations, revealing previously inaccessible structural features of AZs in CNS neurons. Cell-type specific labeling of other endogenous synaptic proteins, such as vesicle-associated proteins, may provide complementary insights to AZ profiling and allow molecular reconstruction of synapses.

Single-cell level analysis of AZ profiles revealed cell-type-specific stereotypy in spatial regulation (Figure 4-6). We found that AZ density in the DPM neurons is consistently high in the α'/β' lobes across individuals (Figure 5). This AZ stereotypy of DPM explains the functional significance of the branches in the α'/β' lobes. Overexpression of a Down syndrome cell adhesion molecule (DScam) variant in the DPM neurons was shown to disrupt their innervation except in the α'/β' lobe, while memory formation remained unaffected (Keene et al., 2006). Similarly, our analysis identified the intracellular stereotypy in AZ distribution of the PPL1- $\alpha 3$ DAN within the $\alpha 3$ compartment (Figure 4). This sub-compartmental organization of the single DAN and layered projections of α/β KCs may underlie the differential dopaminergic modulation of KC subtypes in associative learning (Perisse et al., 2013). While such stereotypy is less pronounced in APL, we found strong correlations between Brp::rGFP intensities of neighboring AZs in contrast to DPM (Figure 6D). This suggests that AZs in APL neurons are organized more locally, allowing them to modulate microcircuits composed of KCs, DANs and MBONs (Amin et al., 2020). Collectively, the spatial scale of AZ organizations depends on cell types, reflecting the function of individual MB neurons.

We observed that Brp::rGFP intensity correlates strongly with the AZ size in KC subtypes, in contrast to DPM and APL neurons (Figure 3). Previous studies using localization microscopy showed that the molecular density of synaptic proteins at AZs can be dynamically adjusted in

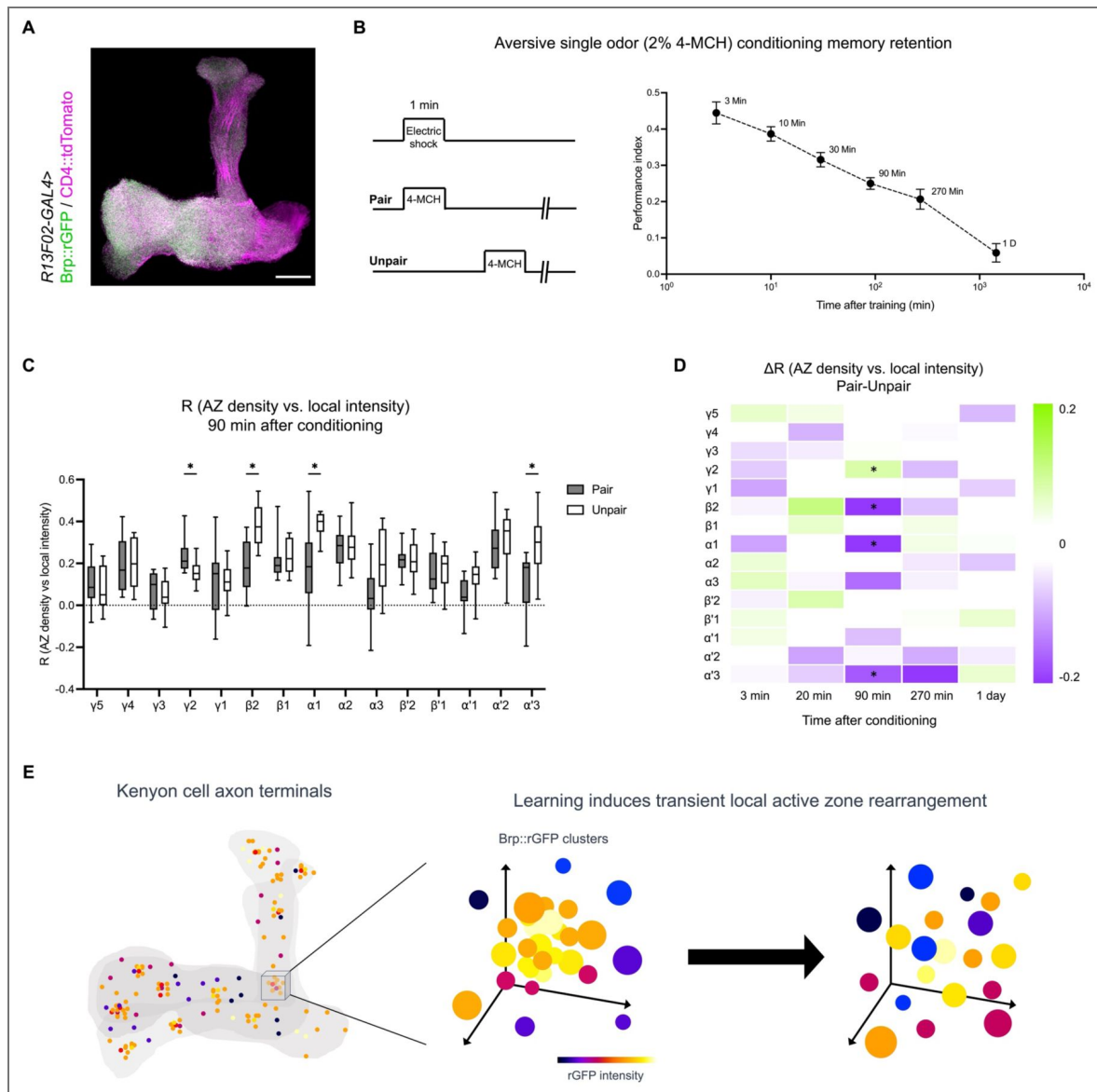


Figure 7. Associative conditioning re-organizes sub-compartmental active zone clusters.

(A) Brp::rGFP (green) and CD4::tdTomato cytoplasm membrane marker (magenta) in KCs, visualized by using *R13F02-GAL4*. (B) Single odor conditioning induces long-lasting memory. Left panel, the experimental design of the aversive single odor conditioning. The paired group receives a concurrent presentation of 2% 4-MCH and 90 V electric shock. The unpaired group first receives the electric shock and then 4-MCH 1 min later. Right panel, preference of *Canton S* between 2% 4-MCH and paraffin oil at 3, 10, 30, 90, 270 minutes and 1 day after single odor conditioning. $n = 8$ for all groups. Error bars show S.E.M. (C) Correlation coefficient (AZ density vs. local intensity of individual Brp::rGFP clusters) of each compartment at 90 min after conditioning. Pair ($n = 11$) vs. Unpair ($n = 12$), γ_2 ($P = 0.0261$), β_2 ($P = 0.0060$), α_1 ($P = 0.0026$), α'_3 ($P = 0.0026$). (D) Heatmap showing the difference of correlation coefficient (AZ density vs. local intensity of Brp::rGFP clusters) between the pair and unpair group at different time points after conditioning. The color indicates the difference and the asterisks in the compartments indicate the significant difference. For 3 min, Pair ($n = 12$), Unpair ($n = 11$); For 20 min, Pair ($n = 12$), Unpair ($n = 11$); For 90 min, Pair ($n = 11$) vs. Unpair ($n = 12$), $P = 0.0247$; For 270 min, Pair ($n = 11$), Unpair ($n = 11$); For 1 D, Pair ($n = 12$), Unpair ($n = 9$); For all results involved statistical comparison, only significant results are shown. Scale bars, 20 μm . Mann-Whitney test with original False Discovery Rate method of Benjamini and Hochberg correction. $*P < 0.05$. Data were presented as box plots showing centre (median), whiskers (Min. to Max.). (E) Schematics showing learning induced local AZ remodeling. Single odor aversive training transiently dismisses high Brp level – high AZ density hot spots in specific compartments of KC terminals.

motor neurons, potentially modulating the synaptic vesicle release (Ghelani et al., 2023; Mrestani et al., 2021; Sachidanandan et al., 2023). The variable Brp concentrations (i.e. the ratio of the Brp::rGFP intensity to the AZ size) in DPM and APL neurons may represent AZs with distinct Brp densities and therefore different functional modes (Figure 3) (Mrestani et al., 2021). Notably, a recent study successfully predicted the neurotransmitter type of neurons using the ultrastructure of dense projections as one of the parameters (Eckstein et al., 2024), suggesting neurotransmitter-specific AZ structures. Interestingly, both the DPM and APL neurons are reported to release multiple neurotransmitters (Davie et al., 2018; Haynes et al., 2015; Lee et al., 2011; Liu and Davis, 2009; Wu et al., 2013; Zeng et al., 2023). Distinct Brp concentrations at AZs might thus support particular release modes or neurotransmitter types.

In KCs, the spatial analysis of local AZ configuration revealed strong correlations between Brp::rGFP intensities at neighboring AZs as well as AZ “hot spots”, regions with high Brp localization and AZ densities (Figure 6). These results suggest previously unidentified, highly localized sub-compartmental AZ structures, which are likely organized across individual KCs. Presynaptic terminals of a single KC are found to be highly heterogeneous in terms of their plasticity upon dopamine modulation and associative learning (Bilz et al., 2020; Davidson et al., 2023). A recent study found the sub-compartmentally heterogeneous activities of dopamine terminals explain memory specificity (Yamagata et al., 2021). Such locally distinct dopamine input may be a source of the AZ hot spot formation in KC terminals. This model is consistent with the localization of two opposing dopamine receptors Dop1R1 and Dop2R to AZs (Hiramatsu et al., 2024).

Furthermore, we showed that the local configuration undergoes structural plasticity upon associative learning that involves presynaptic dopaminergic modulation (Aso et al., 2012, 2010; Aso and Rubin, 2016; Burke et al., 2012; Liu et al., 2012; Yamagata et al., 2015) (Figure 7). Since cAMP signaling plays a crucial role in the AZ structural plasticity especially during associative memory formation (Baltruschat et al., 2021; Sachidanandan et al., 2023; Wu et al., 2025), it may underlie learning-induced plasticity in sub-compartmental AZ configuration (Turrel et al., 2022; Zhang et al., 2018).

Materials and Methods Animals

Flies were maintained on standard cornmeal food at 25 °C under a 12:12 h light-dark cycle for all experiments. All flies used for experiments are 3-7 days old adult males. Flies were transferred to fresh food vials after hatching and flipped every two days before experiments. The GAL4-UAS system was used to express transgenes, and balancers were removed for animals used in all experiments. Fly strains and resources used in this study are as follow: *MB008B-GAL4* (BDSC 68291) (Aso et al., 2014), *MB009B-GAL4* (BDSC 68292) (Aso et al., 2014); *MB370B-GAL4* (BDSC 68319) (Aso et al., 2014), *MB504B-GAL4* (BDSC 68329) (Aso et al., 2014), *VT64246-GAL4* (VDRC 204311) (Tirian and Dickson, 2017), *GH146-GAL4* (BDSC 30026) (Stocker et al., 1997), *R58E02-GAL4* (BDSC 41347) (Pfeiffer et al., 2008), *R13F02-GAL4* (BDSC 48571) (Pfeiffer et al., 2008), *R86E01-GAL4* (BDSC 45914) (Jenett et al., 2012), *Amon-GAL4* (BDSC 30554) (Rhea et al., 2010), *Rh6-GAL4* (BDSC 7464, by Dr. Claude Desplan), *UAS-CD4::tdTomato* (BDSC 35841, by Dr. Yhu Nung Jan & Chun Han), *UAS-Rab3-RNAi* (BDSC 34655) (Perkins et al., 2015), *brp::GFP₁₁* (This study and (Wu et al., 2025)), *UAS-GFP₁₋₁₀* (Kondo et al., 2020).

Generation of *brp::GFP₁₁*

CRISPR/Cas9-mediated homologous recombination was used to insert GFP₁₁ right before the stop codon of *brp*. The donor vector contained a left homology arm of 747 bp, the GFP11 sequence (Kondo et al., 2020), a floxed 3xP3-RFP marker and a right homology arm of 741 bp. A guide RNA sequence (TCGCAAAGCCACAGATACAC) targeting the *brp* locus was cloned into pBfV-U6.2 (Kondo and Ueda, 2013) to generate a gRNA expression vector. The donor and gRNA vectors were co-injected into fertilized eggs carrying the nos-Cas9 transgene (Kondo and Ueda, 2013). Successful transformants were identified by eye-specific RFP fluorescence. Flies were then crossed to a CyO-

Cre balancer to remove the RFP marker. Primers used to clone homology arms are as followed (lowercase letters indicate homologous recombination regions with the doner vector): left arm forward: gcttgatcatgaattcAAGGACATCGAGGAAAAGGAGAAGAAG; left arm reverse: agttggggcgtaggGAAAAAGCTCTTCAAGAAGCCAGCTGGTCC; right arm forward: tagtataggaacttcTCGCAAAGCCACAGA TACACACATCTTGG; right arm reverse: cgggctgcaggaattcATCGTCGATAATTGTAAGTGCATGCTG

qRT-PCR experiment

Procedures are as previously described (Saito et al., 2025 [DOI](#)). For each sample, RNA extract from ~20 heads was purified using TRIzol LS Reagent (Invitrogen) and Direct-zol RNA Microprep kit (ZYMO RESEARCH), following the provided protocol. The concentration of extracted RNA samples was then measured using the NanoDrop Spectrophotometer (Thermo Fisher Scientific) and adjusted to a similar level using diluted water. Reverse transcription was performed using ReverTra Ace qPCR RT Master Mix (TOYOBO, FSQ-301). qRT-PCR was performed using iTaq Universal SYBR Green Supermix (BIO-RAD) and CFX96 Touch Real-Time PCR Detection System (BIO-RAD). No template controls were also prepared and measured alongside. Three primer pairs for *brp* were designed and used in the experiment but only one pair that had Ct<30 for all samples was used. Primers for *brp*, forward: AGCTCAAGGACCACATGGACATC; reverse: GCGCCATATCCACCTGGTTGTC (Sigma-Aldrich). Primers for *Ubiquitin-5E*, forward: TCTTCACTGGTCTGCGTC; reverse: ATGGCTCGACCTCCAAAGTG (eurofins Genomics). Primers for *αTubulin84B*, forward: GATCGTGTCC TCGATTACCGC; reverse: GGGAAGTGAATACGTGGGTAGG (eurofins Genomics).

Light exposure experiment

Procedures are as previously described (Sugie et al., 2015 [DOI](#)). In brief, animals hatched within one day were transferred to constant darkness (DD, 0 LUX) or constant light (LL, 2000-4000 LUX by LED panels) environment at 25 °C and kept for 2-3 days. Animals were then dissected and Brp::rGFP cluster number in individual R8 photoreceptor terminals in the medulla was quantified.

Associative conditioning

For experiments in Figure 7C-D [DOI](#), adult flies at 3-7 days after eclosion were kept in a 25 °C incubator before and after conditioning. Approximately 50 flies were put in a training tube that is set inside the climate box for 2 minutes before the conditioning. Temperature in the climate box was set at 24 °C and the relative humidity was set to 70%. For the paired group, 2% 4-methylcyclohexanol (4-MCH, Sigma-Aldrich, diluted in paraffin oil, Sigma-Aldrich) was presented in a cup with the diameter of 5 mm, together with 12 pulses of electric shocks at 90 V for 1 minute inside the training tube. The unpaired group received the same 1-minute electric shock first, followed by 1-minute rest, and then the same 1-minute 4-MCH presentation. Flies were dissected at 3 min, 20 min, 90 min, 270 min and 1 day after conditioning. Experiments were performed multiple times on different days for a single group.

For measuring the memory decay curve of single odor conditioning, 2% 4-MCH and pure paraffin oil were used for the training. One odor was presented with electric shocks for 1 minute, followed by 1 minute rest, then another odor was presented without electric shocks. For each reciprocal, preference was tested between 2% 4-MCH and paraffin oil. For the differential odor conditioning and short-term memory performance test, 10% 4-MCH and 10% 3-OCT were used for the conditioning, following the same training protocol. Short-term memory performance was tested immediately after the conditioning.

Performance index was calculated as $\#(CS-) - \#(CS+) / (\#(CS-) + \#(CS+))$.

Sample preparation

Animals used in experiments involving comparisons between control and experimental groups were dissected on the same day. Data were collected from multiple batches of experiments performed on different days. Flies were anesthetized by ice and placed on ice before dissection. The dissection was performed in cold PBS to collect the brain without the ventral nerve cord. The brains were subsequently fixed by 2% paraformaldehyde for 1 h and then washed 3×20 min by

0.1% PBT (0.1% Triton X-100 in PBS) in a PCR tube, typically 5-6 brains in one tube. Samples were mounted on microscope slides using mounting medium SeeDB2S (Ke et al., 2016). For nc82 immunostaining, fixed brains were washed 3×20 min with 0.1% PBT and incubated in 3% normal goat serum (NGS; Sigma-Aldrich; G9023)-0.1% PBT blocking solution for 1 hour at room temperature. The nc82 antibody (DSHB) solution was diluted 1:20 in the blocking solution. Samples were incubated in antibody solutions at 4°C for 48 hours for both primary and secondary antibodies.

Confocal imaging

The image acquisition was performed using an Olympus FV1200 confocal microscope platform equipped with both PMT and GaAsP high sensitivity detectors and a 60×/1.42 NA oil immersion objective (PLAPON60XO, Olympus). The Kalman filter was turned on to 2× averaging while scanning. For imaging Brp::rGFP clusters in MB-projecting neurons, scanning speed was set at 2~4 μs/pixel with a voxel size of lateral 0.079 × 0.079 × axial 0.370 μm. For experimental point spread function (PSF) imaging, SeeDB2S immersed beads were scanned in a setting that is 60×/1.42 NA oil immersion objective; 473 nm laser power: 2.0%; 559 nm laser power: 1.5%; voxel size: 0.079 μm × 0.079 μm × 0.370 μm; scanning speed: 4 μs/pixel, to produce multiple bead images.

PSF acquisition and image deconvolution

Sub-resolution fluorescent beads (Tetraspeck Microspheres 0.1 μm, Thermo Fisher Scientific, T7279) were imaged to generate the PSF. The bead solution was diluted with distilled water and sonicated several times to eliminate aggregation. PSF images acquired were processed using Amira software (Thermo Fisher Scientific) with the Extract Point Spread Function module. The PSFs extracted from each image were averaged to create a single PSF for later image deconvolution (Resizing voxel size: 0.079 μm × 0.079 μm × 0.370 μm; 32 pixels × 32 pixels × 21 slices of image size). Image deconvolution was performed by using the Richardson-Lucy iterative non-blind algorithm in a Fiji plugin DeconvolutionLab2 (Sage et al., 2017), or by CLIJ2 GPU-based Richardson-Lucy deconvolution (Haase et al., 2020). Images were deconvolved 20 times iteratively to improve image quality.

Brp clusters segmentation and data analysis

For Brp::rGFP cluster detection, the image stack size was adjusted to double the XY plate pixel number using Fiji to increase the detection precision. The images were subsequently processed using the “3D maxima finder” and the “3D spot segmentation” function of the “3D suite” package in Fiji. 3D maxima were identified for each cluster and used as the starting point for pixel clustering (Ollion et al., 2013). An intensity threshold (threshold) was first set to reduce noise. Pixels having values below the threshold would become NaN (not a number). The “noise tolerance” was then set in the 3D maxima finder to adjust its sensitivity. To be noticed, threshold is only applied to images processed by the 3D maxima finder to generate the “peak” image stack. 3D spot segmentation function uses both the “peak” image and the deconvolved Brp::rGFP image to generate “3D ROIs”. 3D ROIs were then applied on the raw images to quantify the volume (total voxel number in ROI) and Brp::rGFP intensity (total pixel value in ROI) of individual clusters using the “3D ROI manager”. For processing most images, both threshold and noise tolerance were set to 200~400, where the segmentation was optimal based on the F-score benchmarking. “Local thresholding method” in 3D spot segmentation was set to “Gaussian fit”, and “SD value” was set to 2.

The above procedures may include a tiling and stitching step for images and data because of the limited computer power. Typically, an image was tiled into 25 sub-stacks with overlapped margins. After being processed separately, the data acquired from all tiled sub-stacks was stitched back with a deletion of overlapping data points to generate the complete detection result. Nearest neighbors were identified for each particle and spatial distances in between (NND) were calculated using Python. For each sample, mean NND was calculated separately. For a particular Brp::rGFP cluster, the number of surrounding clusters within r ($r = 3 \times \text{mean NND}$) spherically including itself is

referred as the “AZ density”. The “local intensity” of a particular Brp::rGFP cluster is the average of Brp::rGFP intensity of all clusters surrounding within r ($r = 3 \times \text{mean NND}$) including itself. The plot function of Python was used to plot the color-coded 3D reconstructions. A sample code for analyzing NND, AZ density and local intensity is provided with a demonstration data sheet (Supplementary file 1 and analysis code).

Detection optimization

The 3D detection parameters were optimized by matching the 3D maxima detection results to the ground truths. Ground truths were defined by annotating AZs manually, according to the size, position and signal pattern of Brp clusters in the images. The F-score was used to assess detection precision. F-score is the harmonic mean of a system’s precision and recall value, calculated by the following formula: $2 \times [(\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})]$. The precision is the number of true positives divided by the total number of identified positives, including incorrect identification. The recall is the number of true positives divided by the total number of true data points defined as the ground truth. F-scores shown in Figure S5 were automatically calculated using a customized Fiji Marco that aligned 3D maxima with ground truth. Different combinations of noise threshold and noise tolerance were tested for the test images cropped from raw images.

Statistical analysis

Statistical analysis was performed using GraphPad Prism version 8, 9 and 10 or Python. For statistical comparisons, Kruskal-Wallis test with original False Discovery Rate method of Benjamini and Hochberg correction was applied unless otherwise specified. The desired false discovery rate was set to 0.05. Pearson’s correlation coefficient (R) was calculated using Python.

Figure supplements

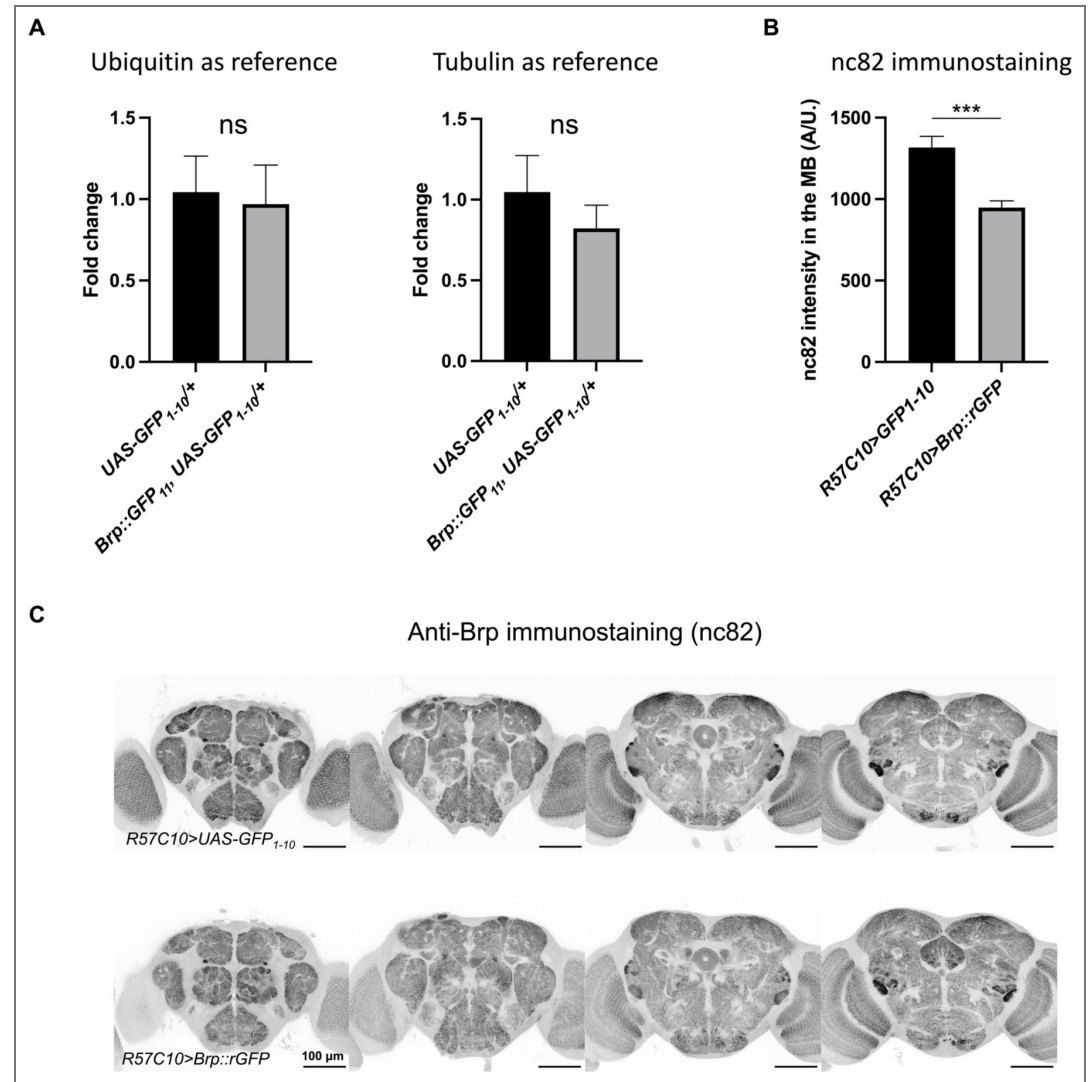


Figure 1 – figure supplement 1. Split-GFP tagging does not affect Brp neuronal expression specificity and expression pattern in the brain. (A) Fold change of *brp* mRNA level against reference genes, measured by qRT-PCR. mRNA of *Ubiquitin-5E* and *αTubulin84B* were used as the reference. $n = 3$ for all groups. Error bars show S.E.M. ns = non-significant. $***p \leq 0.001$. Mann-Whitney test. (B) Anti-Brp (nc82) immunostaining signal in the MB of flies with or without GFP₁₁ insertion. Both have expressions of GFP₁₋₁₀ by *R57C10-GAL4*. $n = 11$ for both groups. Error bars show S.E.M. (C) Anti-Brp immunostaining of brains of flies with or without pan-neuronal Brp::rGFP tagging using *R57C10-GAL4*. Different planes of the image stack were shown. Scale bars, 100 μm.

Figure 1 – figure supplement 2. Neuron-specific assembly of Brp::rGFP.

Brp::rGFP and CD4::tdTomato were visualized in astrocyte-like glia using *R86E01-GAL4*, and in neuropeptide releasing neurons using *Amon-GAL4*, respectively. Images were acquired using comparable settings. Scale bars, 50 μ m.

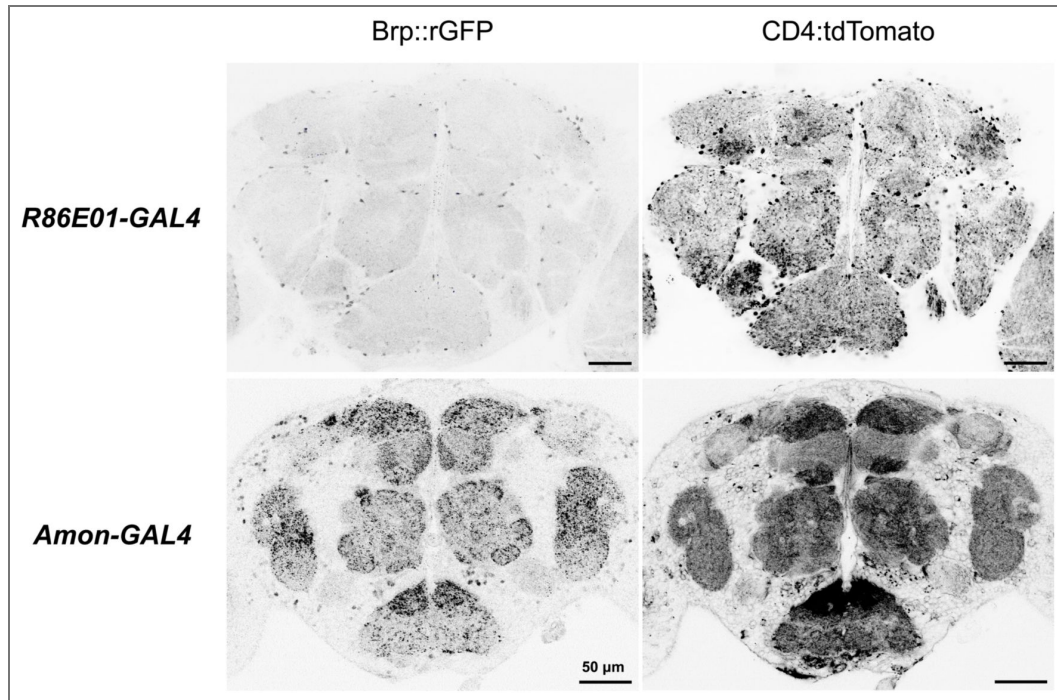
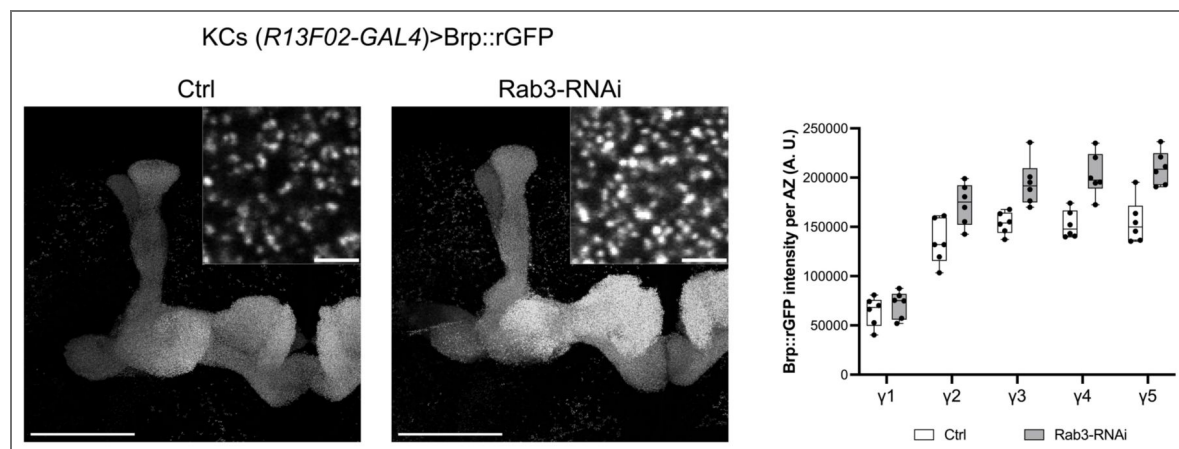


Figure 1 – figure supplement 3. Rab3 knock-down in KCs increases Brp::rGFP intensity.

Rab3 was knocked-down in KCs using RNAi by *R13F02-GAL4*. The median of Brp::rGFP intensity of individual clusters was calculated for each γ compartment. $n = 6$ for both groups. Scale bars, 50 μ m in the overview, 2 μ m in the insets.



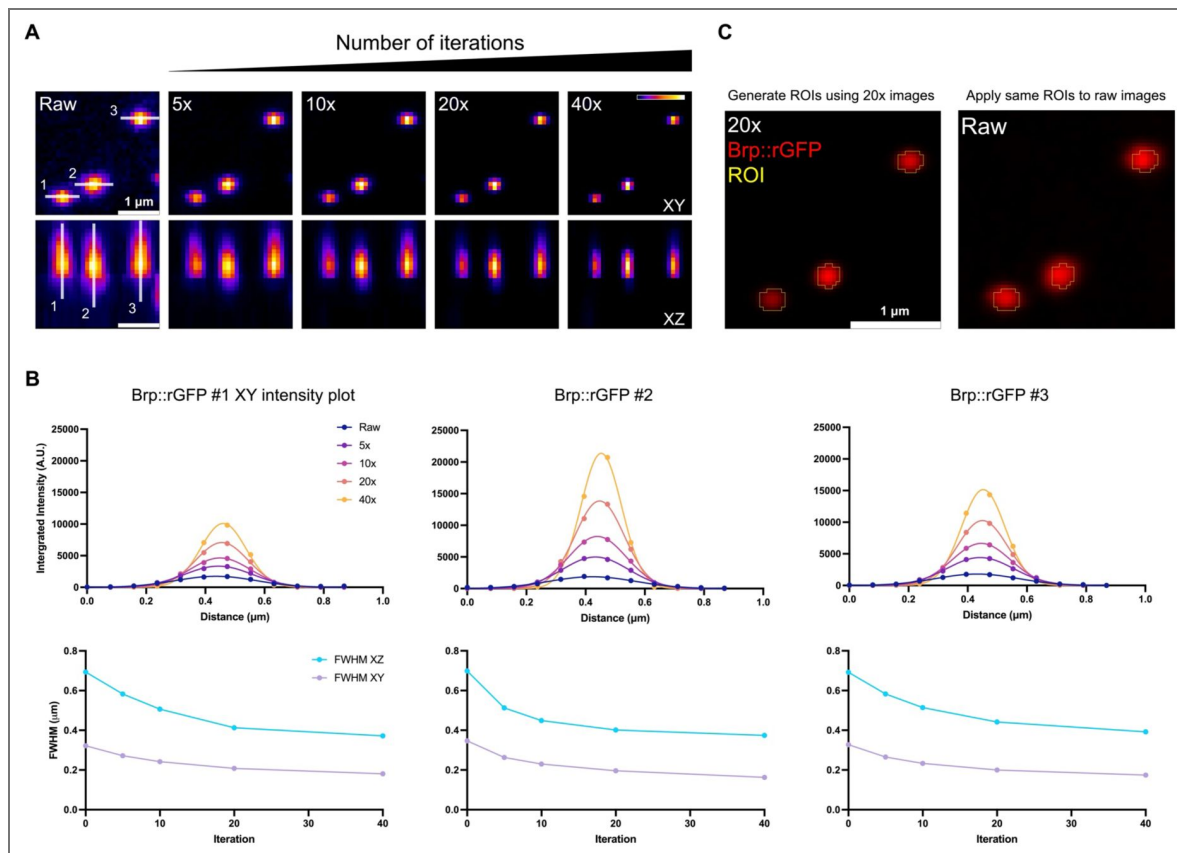


Figure 1 – figure supplement 4. Effect of iterative Richardson-Lucy image deconvolution.

(A) Brp::rGFP clusters of a DPM neuron in raw image, and in images deconvolved for 5–40 iterations. Max projections of the XY and XZ planes were shown. White transparent lines indicate pixels where the intensity profiles were plotted in B. Numbers denote different Brp::rGFP clusters. Scale bars, 1 μm . (B) Upper panels show intensity profiles of Brp::rGFP clusters undergone different iterations of deconvolution. Lower panels show the full width at half maximum (FWHM) value of intensity profiles. (C) 3D ROIs were generated using deconvolved images and then were used to analyze the volume and intensity of individual Brp::rGFP clusters in raw images. Scale bars, 1 μm .

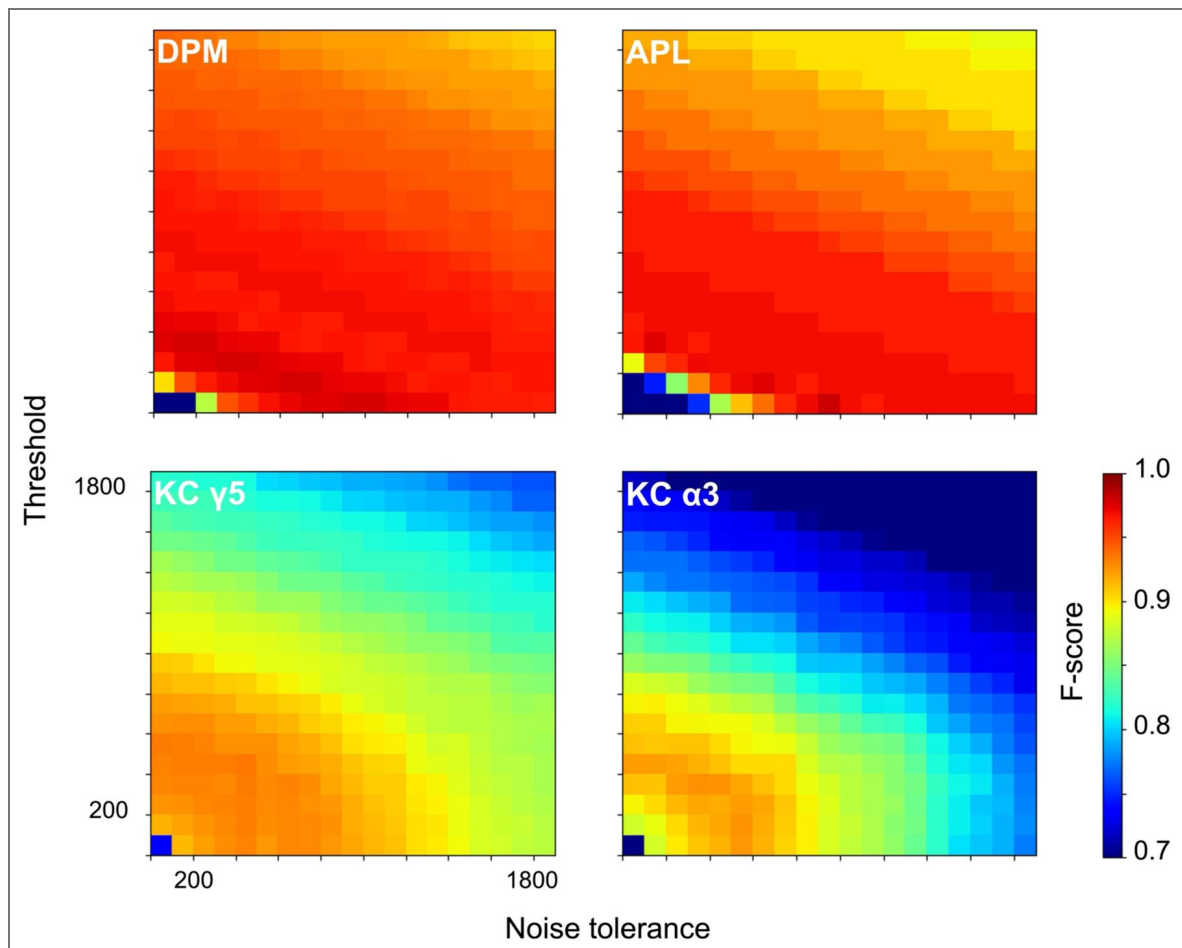


Figure 1 – figure supplement 5. Heatmaps of F-scores showing the performance of the pipeline in detecting individual Brp::rGFP clusters in different cell types.

An intensity threshold (threshold) is applied to reduce background noise when detecting 3D maxima. Noise tolerance adjusts the sensitivity of 3D maxima detection. Color represents F-score value.

Figure 7 – figure supplement 1. Gross anatomy of KC terminals is not affected by Brp::rGFP tagging.

Morphology of KC terminals in flies with or without GFP₁₁ insertion. Both groups express GFP₁₋₁₀ and CD4::tdTomato, driven by *R13F02-GAL4*. Scale bars, 20 μ m.

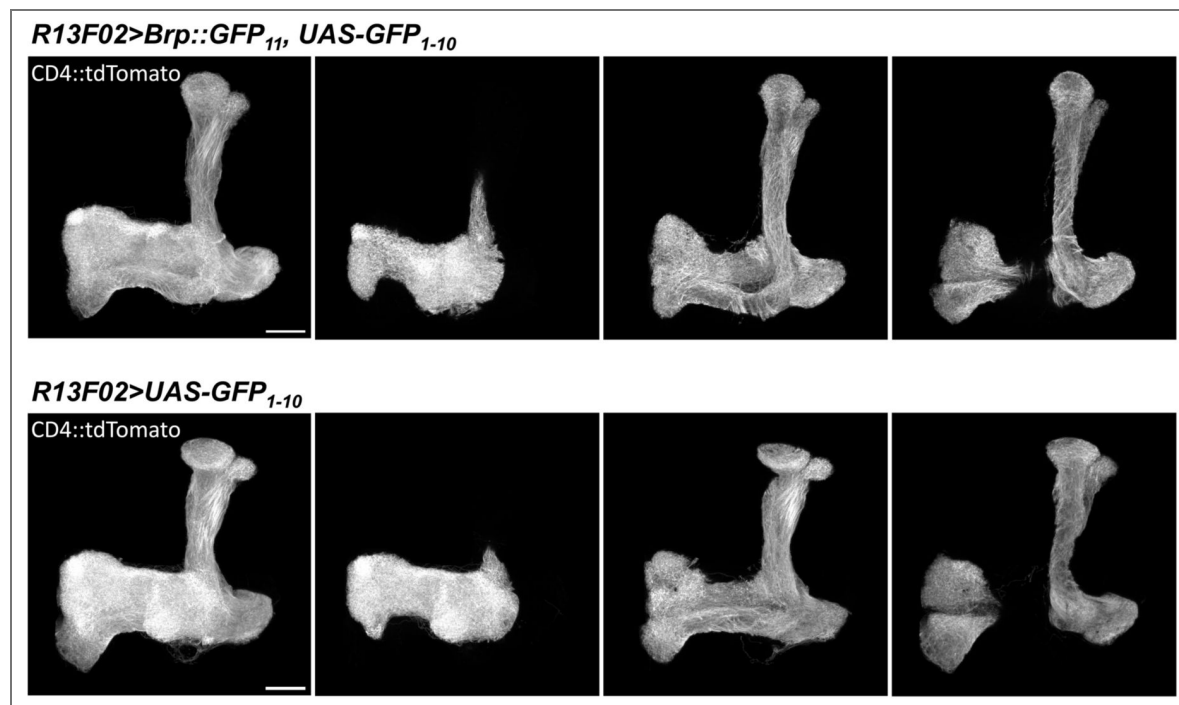
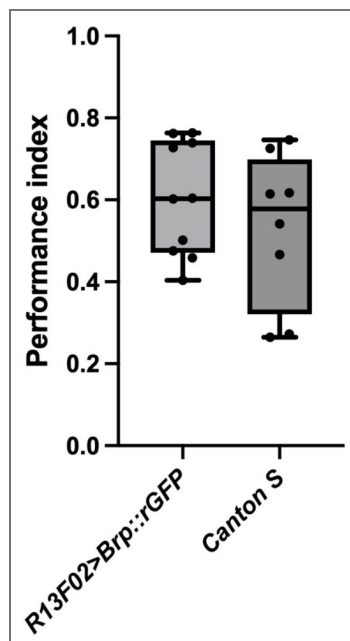


Figure 7 – figure supplement 2. Brp::rGFP tagging in KCs does not affect short-term memory performance.

10% 4-MCH or 10% 3-OCT was paired with 90 V electric shocks. Odor preference was tested immediately after conditioning. GFP reconstitution was induced in KCs using *R13F02-GAL4*. *R13F02>Brp::rGFP*, n = 10; *Canton S*, n = 8. ns = non-significant. Welch's t test.



Acknowledgements

We thank all lab members of Tanimoto Lab at Tohoku University for valuable discussions. We thank Dr. Yuki Tsukada at Keio University for crucial advice on Fiji-based image analysis. We thank Vienna Drosophila Resource Center and Bloomington Drosophila Stock Center for transgenic flies.

Additional information

Data and materials availability

Further information and requests on resources and reagents should be directed to and will be fulfilled by the corresponding author, Dr. Hiromu Tanimoto (hiromut@m.tohoku.ac.jp). All data and codes needed to evaluate the conclusions in the paper are present in the paper.

Author contributions

Conceptualization: HW, HT Methodology: HW, YM, NY, SE, SK Investigation: HW, YM, SE Visualization: HW, SE Resources: SK Supervision: HW, NY, HT Writing—original draft: HY, HT Writing—review & editing: HW, SE, HT

Funding

Funder	Grant reference number	Author
MEXT Japan Science and Technology Agency (JST)	JPMJSP2114	Hongyang Wu
Tohoku University (東北大学)	Frontier Research in Duo	Hiromu Tanimoto
Ministry of Education, Culture, Sports, Science and Technology (MEXT)	20H00519	Hiromu Tanimoto
Ministry of Education, Culture, Sports, Science and Technology (MEXT)	22H05481	Hiromu Tanimoto
Ministry of Education, Culture, Sports, Science and Technology (MEXT)	22KK0106	Hiromu Tanimoto

Author ORCID iDs

Hongyang Wu: <https://orcid.org/0000-0001-7889-6941>

Additional files

Supplementary file 1. [Python code used for nearest neighbor, local intensity and AZ density analysis.](#)

Supplementary file 2. [Example data of a DPM neuron generated by the 3D spot segmentation, before being processed by the analysis code.](#)

Supplementary file 3. [Numerical data for figures.](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

The study by Wu et al. uses endogenous bruchpilot expression in a cell-type-specific manner to assess synaptic heterogeneity in adult *Drosophila melanogaster* mushroom body output neurons. The authors performed genomic on locus tagging of the presynaptic scaffold protein bruchpilot (brp) with one part of splitGFP (GFP11) using the CRISPR/Cas9 methodology and co-expressed the other part of splitGFP (GFP1-10) using the GAL4/UAS system. Upon expression of both parts of splitGFP, fluorescent GFP is assembled at the C-terminus of brp, exactly where brp is endogenously expressed in active zones. For manageable analysis, a high-throughput pipeline was developed. This analysis evaluated parameters like location of brp clusters, volume of clusters, and cluster intensity as a direct measure of the relative amount of brp expression levels on site using publicly available 3D analysis tools that are integrated in Fiji. Analysis was conducted for different mushroom body cell types in different mushroom body lobes using various specific GAL4 drivers. Further validation was provided by extending analysis to R8 photoreceptors that reside in the fly medulla. To test this new method of synapse assessment, Wu et al. performed an associative learning experiment in which an odor was paired with an aversive stimulus and found that in a specific time frame after conditioning, the new analysis solidly revealed changes in brp levels at specific synapses that are associated with aversive learning. Additionally, brp levels were assessed in R8 photoreceptor terminals upon extended exposure to light.

Strengths:

Expression of splitGFP bound to brp enables intensity analysis of brp expression levels as exactly one GFP molecule is expressed per brp. This is a great tool for synapse assessment. This tool can be widely used for any synapse as long as driver lines are available to co-express the other part of splitGFP in a cell-type-specific manner. As neuropils and thus brp

label can be extremely dense, the analysis pipeline developed here is very useful and important. The authors have chosen an exceptionally dense neuropil - the mushroom bodies - for their analysis and compellingly show that brp assessment can be achieved even with such densely packed active zones. The result that brp levels change upon associative learning in an experiment with odor presentation paired with punishment is likewise compelling and strongly suggests that the tool and pipeline developed here can be used in an *in vivo* context. Thus, the tool and its uses have the potential to fundamentally advance protein analysis not only at the synapse but especially there.

Weaknesses:

The weaknesses I perceived originally were satisfactorily explained and refuted.

<https://doi.org/10.7554/eLife.107663.2.sa3>

Reviewer #2 (Public review):

Summary:

The authors developed a cell-type-specific fluorescence-tagging approach using a CRISPR/Cas9 induced split-GFP reconstitution system to visualize endogenous Bruchpilot (BRP) clusters at presynaptic active zones (AZ) in specific cell types of the mushroom body (MB) in the adult *Drosophila* brain. This AZ profiling approach was implemented in a high-throughput quantification process allowing to compare synapse profiles within single cells, cell-types, MB compartments and between different individuals. Aim is to in more detail analyze neuronal connectivity and circuits in this center of associative learning, notoriously difficult to investigate due to the density of cells and structures within the cells. The authors detect and characterize cell-type specific differences in BRP-dependent profiling of presynapses in different compartments of the MB, while intracellular AZ distribution was found to be stereotyped. Next to the descriptive part characterizing various AZ profiles in the MB, the authors apply an associative learning assay and Rab3 knock-down and detected consequent AZ reorganization.

Strengths:

The strength of this study lies in the outstanding resolution of synapse profiling in the extremely dense compartments of the MB. This detailed analysis will serve as an entry point for many future studies of synapse diversity in connection with functional specificity to uncover the molecular mechanisms underlying learning and memory formation and neuronal network logic. Therefore, this approach is of high importance to the scientific community and represents a valuable tool to investigate and correlate AZ architecture and synapse function in the CNS.

Weaknesses:

The results and conclusions presented in this study are conclusively and well supported by the data presented and appropriate controls. As a comment that could possibly aid and strengthen the manuscript (but not required for acceptance of the manuscript): The experiments in the study are based on split-GFP lines (BRP:GFP11 and UAS-GFP1-10). The authors clearly validate the new on-locus construct with a genomic GFP insertion (qPCR, confocal and STED imaging of the brain with anti-BRP (Nc82), MB morphology and memory formation). It would be important to comment on the significant overall intensity decrease of anti-BRP (Nc82) in Fig. S1B (R57C10>BRP::rGFP) and possibly a Western Blot with a correlative antibody staining against BRP might help to show that BRP protein level are not affected. Additionally, it would be important to state, at least in the Materials and Methods

section, that the flies are not homozygous viable (and to offer an explanation) and to state that all experiments were performed with heterozygous flies.

<https://doi.org/10.7554/eLife.107663.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors develop a tool for marking presynaptic active zones in *Drosophila* brains, dependent on the GAL4 construct used to express a fragment of GFP, which will incorporate with a genome-engineered partial GFP attached to the active zone protein bruchpilot - signal will be specific to the GAL4 expressing neuronal compartment. They then use various GAL4s to examine innervation onto the mushroom bodies to dissect compartment specific differences in the size and intensity of active zones. After a description of these differences, they induce learning in flies with classic odour/electric shock pairing and observe changes after conditioning that are specific to the paired conditioning/learning paradigm.

Strengths:

The imaging and analysis appears strong. The tool is novel and exciting.

Weaknesses:

I feel that the tool could do with a little more characterisation. It is assumed that the puncta observed are AZs with no further definition or characterisation. It is not resolved if the AZs visualised here simply tagged, or are the constructs incorporated to be an active functional part of the AZ.

Comments on revisions:

Apologies, I should have thought of this in the first round of review. An experiment I would suggest (and it is not a difficult one) to address the functionality of the marker: It is mentioned that the genetically tagged half of the construct is homozygous lethal. Can this be placed in trans to a *brp* null, with a neuronal UAS-expression of the other half of Brp-GFP - Are the animals then 1) alive, and 2) able to fly (*brp* mutants can't fly, hence the name 'crashpilot') - a rescue would suggest (and that is all that would be needed here) that the reconstituted *brp*-GFP has function.

On another note, the paper keeps switching between different DAN-GAL4 lines. In 1H, 2Band 4A, there are informative cartoons showing the extension of the neurons for PPL1, APL and DPM neurons - could these be incorporated into figures 5, 6 and 7, and the supplementary figures to help orient the reader. Ideally they would refer to a figure (in Fig 1?) -to refer to the groups of DANs in the adult brain that are known to innervate the MBs (e.g. Fig1 in Mao and Davis, Front in Neural Circuits 2009). I suggest this because I feel that this tool will be widely used, and if non-MB aficionados can follow what's being done here I feel it will be more widely accepted.

<https://doi.org/10.7554/eLife.107663.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The study by Wu et al. uses endogenous bruchpilot expression in a cell-type-specific manner to assess synaptic heterogeneity in adult Drosophila melanogaster mushroom body output neurons. The authors performed genomic on locus tagging of the presynaptic scaffold protein bruchpilot (BRP) with one part of splitGFP (GFP11) using the CRISPR/Cas9 methodology and co-expressed the other part of splitGFP (GFP1-10) using the GAL4/UAS system. Upon expression of both parts of splitGFP, fluorescent GFP is assembled at the N-terminus of BRP, exactly where BRP is endogenously expressed in active zones. For manageable analysis, a high-throughput pipeline was developed. This analysis evaluated parameters like location of BRP clusters, volume of clusters, and cluster intensity as a direct measure of the relative amount of BRP expression levels on site, using publicly available 3D analysis tools that are integrated in Fiji. Analysis was conducted for different mushroom body cell types in different mushroom body lobes using various specific GAL4 drivers. To test this new method of synapse assessment, Wu et al. performed an associative learning experiment in which an odor was paired with an aversive stimulus and found that, in a specific time frame after conditioning, the new analysis solidly revealed changes in BRP levels at specific synapses that are associated with aversive learning.

Strengths:

Expression of splitGFP bound to BRP enables intensity analysis of BRP expression levels as exactly one GFP molecule is expressed per BRP. This is a great tool for synapse assessment. This tool can be widely used for any synapse as long as driver lines are available to co-express the other part of splitGFP in a cell-type-specific manner. As neuropils and thus the BRP label can be extremely dense, the analysis pipeline developed here is very useful and important. The authors have chosen an exceptionally dense neuropil - the mushroom bodies - for their analysis and convincingly show that BRP assessment can be achieved with such densely packed active zones. The result that BRP levels change upon associative learning in an experiment with odor presentation paired with punishment is likewise convincing, and strongly suggests that the tool and pipeline developed here can be used in an in vivo context.

Weaknesses:

Although BRP is an important scaffold protein and its expression levels were associated with function and plasticity, I am still somewhat reluctant to accept that synapse structure profiling can be inferred from only assessing BRP expression levels and BRP cluster volume. Also, is it guaranteed that synaptic plasticity is not impaired by the large GFP fluorophore? Could the GFP10 construct that is tagged to BRP in all BRP-expressing cells, independent of GAL4, possibly hamper neuronal function? Is it certain that only active zones are labeled? I do see that plastic changes are made visible in this study after an associative learning experiment with BRP intensity and cluster volume as read-out, but I would be reassured by direct measurement of synaptic plasticity with splitGFP directly connected to BRP, maybe at a different synapse that is more accessible.

We appreciate the reviewer's comments. In the revised manuscript, we have clarified that Brp is an important, but not the only player in the active zone. We have included new data to demonstrate that split-GFP tagging does not severely affect the localization and plasticity of Brp and the function of synapses by showing: (1) nanoscopic localization of Brp::rGFP using STED imaging; (2) colocalization between Brp::rGFP and anti-Brp signals/VGCCs; (3) activity-dependent Brp remodeling in R8 photoreceptors; (4) no defect in memory performance when labeling Brp::rGFP in KCs; These four lines of additional evidence further corroborate our approach to characterize endogenous Brp as a proxy of active zone structure.

Reviewer #2 (Public review):

Summary:

The authors developed a cell-type specific fluorescence-tagging approach using a CRISPR/Cas9 induced split-GFP reconstitution system to visualize endogenous Bruchpilot (BRP) clusters as presynaptic active zones (AZ) in specific cell types of the mushroom body (MB) in the adult Drosophila brain. This AZ profiling approach was implemented in a high-throughput quantification process, allowing for the comparison of synapse profiles within single cells, cell types, MB compartments, and between different individuals. The aim is to analyse in more detail neuronal connectivity and circuits in this centre of associative learning. These are notoriously difficult to investigate due to the density of cells and structures within a cell. The authors detect and characterize cell-type-specific differences in BRP-dependent profiling of presynapses in different compartments of the MB, while intracellular AZ distribution was found to be stereotyped. Next to the descriptive part characterizing various AZ profiles in the MB, the authors apply an associative learning assay and detect consequent AZ re-organisation.

Strengths:

The strength of this study lies in the outstanding resolution of synapse profiling in the extremely dense compartments of the MB. This detailed analysis will be the entry point for many future analyses of synapse diversity in connection with functional specificity to uncover the molecular mechanisms underlying learning and memory formation and neuronal network logics. Therefore, this approach is of high importance for the scientific community and a valuable tool to investigate and correlate AZ architecture and synapse function in the CNS.

Weaknesses:

The results and conclusions presented in this study are, in many aspects, well-supported by the data presented. To further support the key findings of the manuscript, additional controls, comments, and possibly broader functional analysis would be helpful. In particular:

(1) All experiments in the study are based on split-GFP lines (BRP:GFP11 and UAS-GFP1-10). The Materials and Methods section does not contain any cloning strategy (gRNA, primer, PCR/sequencing validation, exact position of tag insertion, etc.) and only refers to a bioRxiv publication. It might be helpful to add a Materials and Methods section (at least for the BRP:GFP11 line). Additionally, as this is an on locus insertion the in BRP-ORF, it needs a general validation of this line, including controls (Western Blot and correlative antibody staining against BRP) showing that overall BRP expression is not compromised due to the GFP insertion and localizes as BRP in wild type flies, that flies are viable, have no defects in locomotion and learning and memory formation and MB morphology is not affected compared to wild type animals.

We thank the reviewer for suggesting these important validations. We included details of the design of the construct and insertion site to the Methods section, performed several new experiments to validate the split-GFP tagging of Brp, and present the data in the revision.

First, to examine whether the transcription of the *brp* gene is unaffected by the insertion of GFP₁₁, we conducted qRT-PCR to compare the *brp* mRNA levels between *brp::GFP₁₁*, *UAS-GFP1-10* and *UAS-GFP1-10* and found no difference (Figure 1 - figure supplement 1A).

To further verify the effect of GFP₁₁ tagging at the protein level, we performed anti-Brp (nc82) immunohistochemistry of brains where GFP is reconstituted pan-neuronally. We found

unaltered neuropile localization of nc82 signals (Figure 1 - figure supplement 1C). In presynaptic terminals of the mushroom body calyx, we found integration of Brp::rGFP to nc82 accumulation (Figure 1D). We performed super-resolution microscopy to verify the configuration of Brp::rGFP and confirmed the donut-shape arrangement of Brp::rGFP in the terminals of motor neurons (see Wu, Eno et al., 2025 PLOS Biology), corroborating the nanoscopic assembly of Brp::rGFP at active zones (Kittel et al., 2006 Science).

Furthermore, co-expression of RFP-tagged voltage-gated calcium channel alpha subunit Cacophony (Cac) and Brp::rGFP in PAM-γ5 dopaminergic neurons revealed strong presynaptic colocalization of their punctate clusters (Figure 1E), suggesting that rGFP tagging of Brp did not damage key protein assembly at active zones (Kawasaki et al., 2004 J Neuroscience; Kittel et al., Science).

These lines of evidence suggest that the localization of endogenous Brp is barely affected by the C-terminal GFP₁₁ insertion or GFP reconstitution therewith. This is in line with a large body of studies confirming that the N-terminal region and coiled-coil domains, but not the C-terminal, region of Brp are necessary and sufficient for active zone localization (Fouquet et al., 2009 J Cell Biol; Oswald et al., 2010 J Cell Biol; Mosca and Luo, 2014 eLife; Kiragasi et al., 2017 Cell Rep; Akbergenova et al., 2018 eLife; Nieratschker et al., 2009 PLoS Genet; Johnson et al., 2009 PLoS Biol; Hallermann et al., 2010 J Neurosci). We nevertheless report homozygous lethality and found the decreased immunoreactive signals in flies carrying the GFP₁₁ insertion (Figure 1 - figure supplement 1B).

For these reasons, we always use heterozygotes for all the experiments therefore there is no conspicuous defect in locomotion as reported in the original study (Wagh et al., 2005 Neuron). To functionally validate the heterozygotes, we measured the aversive olfactory memory performance of flies where GFP reconstitution was induced in Kenyon cells using *R13F02-GAL4*. We found that all these transgenes did not alter mushroom body morphology (Figure 7 - figure supplement 1) or memory performance as compared to wild-type flies (Figure 7 - figure supplement 2), suggesting the synapse function required for short-term memory formation is not affected by split-GFP tagging of Brp.

(2) Several aspects of image acquisition and high-throughput quantification data analysis would benefit from a more detailed clarification.

(a) For BRP cluster segmentation it is stated in the Materials and Methods state, that intensity threshold and noise tolerance were "set" - this setting has a large effect on the quantification, and it should be specified and setting criteria named and justified (if set manually (how and why) or automatically (to what)). Additionally, if Python was used for "Nearest Neighbor" analysis, the code should be made available within this manuscript; otherwise, it is difficult to judge the quality of this quantification step.

(b) To better evaluate the quality of both the imaging analysis and image presentation, it would be important to state, if presented and analysed images are deconvolved and if so, at least one proof of principle example of a comparison of original and deconvoluted file should be shown and quantified to show the impact of deconvolution on the output quality as this is central to this study.

We thank the reviewer for suggesting these clarifications. We have included more description to the revised manuscript to clarify the setting of segmentation, which was manually adjusted to optimize the F-score (previous Figure 1D, now moved to Figure 1 -figure supplement 5). We have included the code used for analyzing nearest neighbor distance, AZ density and local Brp density in the revised manuscript (Supplementary file 1), together with a pre-processed sample data sheet (Supplementary file 2).

Regarding image deconvolution, we have clarified the differential use of deconvolved and not-deconvolved images in the revised manuscript. We have also included a quantitative

evaluation of Richardson-Lucy iterative deconvolution (Figure 1 - figure supplement 4). We used 20 iterations due to only marginal FWHM improvement beyond this point (Figure 1 - figure supplement 4).

(3) The major part of this study focuses on the description and comparison of the divergent synapse parameters across cell-types in MB compartments, which is highly relevant and interesting. Yet it would be very interesting to connect this new method with functional aspects of the heterogeneous synapses. This is done in Figure 7 with an associative learning approach, which is, in part, not trivial to follow for the reader and would profit from a more comprehensive analysis.

(a) It would be important for the understanding and validation of the learning induced changes, if not (only) a ratio (of AZ density/local intensity) would be presented, but both values on their own, especially to allow a comparison to the quoted, previous AZ remodelling analysis quantifying BRP intensities (ref. 17, 18). It should be elucidated in more detail why only the ratio was presented here.

We thank the reviewer for the suggestion on the presentation of learning-induced Brp remodeling. The reported values in Figure 7C are the correlation coefficient of AZ density and local intensity in each compartment, but not the ratio. These results suggest that subcompartment-sized clusters of AZs with high Brp accumulation (Figure 6) undergo local structural remodeling upon associative learning (Figure 7). For clarity, we have included a schematic of this correlation and an example scatter plot to Figure 6. Unlike the previous studies (refs 17 and 18), we did not observe robust learning-dependent changes in the Brp intensity, possibly due to some confounding factors such as overall expression levels and conditioning protocols as described in the previous and following points, respectively.

(b) The reason why a single instead of a dual odour conditioning was performed could be clarified and discussed (would that have the same effects?).

(c) Additionally, "controls" for the unpaired values - that is, in flies receiving neither shock nor odour - it would help to evaluate the unpaired control values in the different MB compartments.

We use single odor conditioning because it is the simplest way to examine the effect of odor-shock association by comparing the paired and unpaired group. Standard differential conditioning with two odors contains unpaired odor presentation (CS-) even in the 'paired' group. We now show that single-odor conditioning induces memory that lasts one day as in differential conditioning (Figure 7B; Tully and Quinn, J Comp Phys A 1985).

(d) The temporal resolution of the effect is very interesting (Figure 7D), and at more time points, especially between 90 and 270 min, this might raise interesting results.

The sampling time points after training was chosen based on approximately logarithmic intervals, as the memory decay is roughly exponential (Figure 7B). This transient remodeling is consistent with the previous studies reporting that the Brp plasticity was short-lived (Zhang et al., 2018 Neuron; Turrel et al., 2022 Current Biol).

(e) Additionally, it would be very interesting and rewarding to have at least one additional assay, relating structure and function, e.g. on a molecular level by a correlative analysis of BRP and synaptic vesicles (by staining or co-expression of SV-protein markers) or calcium activity imaging or on a functional level by additional learning assays.

We thank the reviewer for raising this important point. We have performed calcium imaging of KC presynaptic terminals to correlate the structure and function in another study (see Figure 2 in Wu, Eno et al., 2025 PLOS Biology for more detail). The basal presynaptic calcium

pattern along the γ compartments is strikingly similar to the compartmental heterogeneity of Brp accumulation (see also Figure 2 in this study). Considering colocalization of other active-zone components, such as Cac (Figure 1E), we propose that the learning-induced remodeling of local Brp clusters should transiently modulate synaptic properties.

As a response to other reviewers' interest, we used Brp::rGFP to measure different forms of Brp-based structural plasticity upon constant light exposure in the photoreceptors and upon silencing *rab3* in KCs. Since these experiments nicely reproduced the results of previous studies (Sugie et al., Neuron 2013; Graf et al., Neuron 2009), we believe the learning-induced plasticity of Brp clustering in KCs has a transient nature.

Reviewer #3 (Public review):

Summary:

The authors develop a tool for marking presynaptic active zones in Drosophila brains, dependent on the GAL4 construct used to express a fragment of GFP, which will incorporate with a genome-engineered partial GFP attached to the active zone protein bruchpilot - signal will be specific to the GAL4-expressing neuronal compartment. They then use various GAL4s to examine innervation onto the mushroom bodies to dissect compartment-specific differences in the size and intensity of active zones. After a description of these differences, they induce learning in flies with classic odour/electric shock pairing and observe changes after conditioning that are specific to the paired conditioning/learning paradigm.

Strengths:

The imaging and analysis appear strong. The tool is novel and exciting.

Weaknesses:

I feel that the tool could do with a little more characterisation. It is assumed that the puncta observed are AZs with no further definition or characterisation.

We performed additional validation on the tool, including (1) nanoscopic localization of Brp::rGFP using STED imaging; (2) colocalization between Brp::rGFP and anti-Brp signals/VGCCs (Figure 1D-E); 3) activity-dependent active zone remodeling in R8 photoreceptors (Figure 1F). These will be detailed in our point-by-point response below.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) The authors keep stating, they profile or assess synaptic structure by analyzing BRP localization, cluster volume, and intensity. However, I do not think that BRP cluster volume and intensity warrant an educated statement about presynaptic structure as a whole. I do not challenge the usefulness of BRP cluster analysis for synapse evaluation, but as there are so many more players involved in synaptic function, BRP analysis certainly cannot explain it all. This should at least be discussed.

It is correct that Brp is not the only player in the active zone. We have included more discussion on the specific role of Brp (line 84 to 89) and other synaptic markers (line 250) and edited potentially misunderstanding text.

(2) I do see that changes in BRP expression were observed following associative learning, but is it certain, that synaptic plasticity is generally unaffected by the large GFP fluorophore? BRP is grabbing onto other proteins, both with its C- and N-termini. As the

GFP is right before the stop codon, it should be at the N-terminus. How far could BRP function be hampered by this? Is there still enough space for other proteins to interact?

We thank the reviewer for sharing the concerns. We here provided three lines of evidence to demonstrate that the Brp assembly at active zones required for synaptic plasticity is unaffected by split-GFP tagging.

First, we assessed olfactory memory of flies that have Brp::rGFP labeled in Kenyon cells and found the performance comparable to wild-type (Figure 7 - figure supplement 2), suggesting the Brp function required for olfactory memory (Knapek et al., J Neurosci 2011) is unaffected by split-GFP tagging.

Second, we measured Brp remodeling in photoreceptors induced by constant light exposure (LL; Sugie et al., 2015 Neuron). Consistent with the previous study, we found that LL decreased the numbers of Brp::rGFP clusters in R8 terminals in the medulla, as compared to constant dark condition (DD). This result validates the synaptic plasticity involving dynamic Brp rearrangement in the photoreceptors. We have included this result into the revised manuscript (Figure 1F).

To further validate protein interaction of Brp::rGFP, we focused on Rab3, as it was previously shown to control Brp allocation at active zones (Graf et al., 2009 Neuron). To this end, we silenced *rab3* expression in Kenyon cells using RNAi and measured the intensity of Brp::rGFP clusters in γ Kenyon cells. As previously reported in the neuromuscular junction, we found that *rab3* knock-down increased Brp::rGFP accumulation to the active zones, suggesting that Brp::rGFP represents the interaction with Rab3. We have included all the new data to the revised manuscript (Figure 1 - figure supplement 3).

(3) It may well be that not only active-zone-associated BRP is labeled but possibly also BRP molecules elsewhere in the neuron. I would like to see more validation, e.g., the percentage of tagged endogenous BRP associated with other presynaptic proteins.

To answer to what extent Brp::rGFP clusters represent active zones, we double-labelled Brp::rGFP and Cac::tdTomato (Cacophony, the alpha subunit of the voltage-gated calcium channels). We found that 97% of Brp::rGFP clusters showed co-localization with Cac::tdTomato in PAM- γ 5 dopamine neurons terminals (Figure 1E), suggesting most Brp::rGFP clusters represent functional AZs.

(4) Z-size is ~200 nm, while x/y pixel size is ~75 nm during acquisition. How far down does the resolution go after deconvolution?

The Z-step was 370 nm and XY pixel size was 79 nm for image acquisition. We performed 20 iterations of Richardson-Lucy deconvolution using an empirical point spread function (PSF). We found that the effect of deconvolution on the full-width at half maximum (FWHM) of Brp::rGFP clusters improves only marginally beyond 20 iterations, when the XY FWHM is around 200 nm and the XZ FWHM is around 450 nm (Figure 1 - figure supplement 4).

(5) Figure Legend 7: What is a "cytoplasm membrane marker"? Does this mean membrane-bound tdTom is sticking into the cytoplasm?

We apologize for the typo and have corrected it to "plasma membrane marker".

(6) At the end of the introduction: "characterizing multiple structural parameters..." - which were these parameters? I was under the assumption that BRP localization, cluster volume, and intensity were assessed. I do not see how these are structural parameters. Please define what exactly is meant by "structural parameters".

We apologize for the confusion. By "structural parameters", we indeed referred to the volume, intensity and molecular density of Brp::rGFP clusters. We have revised the sentence

to “Characterizing the distinct parameters and localization of Brp::rGFP cluster.”

(7) Next to last sentence of the introduction: "Characterizing multiple structural parameters revealed a significant synaptic heterogeneity within single neurons and AZ distribution stereotypy across individuals." What do the authors mean by "significant synaptic heterogeneity"?

By “synaptic heterogeneity”, we refer to the intracellular variability of active zone cytomatrices reported by Brp clusters. For instance, the intensities of Brp::rGFP clusters within Kenyon cell subtypes were variable among compartments (Figure 2). Intracellular variability of the Brp concentration of individual active zones was higher in DPM and APL neurons than Kenyon cells (Figure 3). These variabilities demonstrate intracellular synaptic heterogeneity. We have revised the sentence to be more specific to the different characters of Brp clusters.

(8) I do not understand the last sentence of the introduction. "These cell-type-specific synapse profiles suggest that AZs are organized at multiple scales, ranging from neighboring synapses to across individuals." What do the authors mean by "ranging from neighboring synapses to across individuals"? Does this mean that even neighboring synapses in the same cell can be different?

We have revised the sentence to “These cell-type-specific synapse profiles suggest that AZs are spatially organized at multiple scales, ranging from interindividual stereotypy to neighboring synapses in the same cells.”

By “neighboring synapses”, we refer to the nearest neighbor similarity in Brp levels in some cell-types (Figure 6A-C), and also the sub-compartmental dense AZ clusters with high Brp level in Kenyon cells (Figure 6D-H). By “across individuals”, we refer to the individually conserved active zone distribution patterns in some neurons (Figure 5).

(9) The title talks about cell-type-specific spatial configurations. I do not understand what is meant by "spatial configurations"? Do you mean BRP cluster volume? I think the title is a little misleading.

By “spatial configuration”, we refer to the arrangement of Brp clusters within individual mushroom body neurons. This statement is based on our findings on the intracellular synaptic heterogeneity (see also response to comment #7). We have streamlined the text description in the revised manuscript for clarity.

Reviewer #2 (Recommendations for the authors):

(1) For Figure 3A: exemplary two AZs are compared here, a histogram comparing more AZs would aid in making the point that in general, AZ of similar size have different BRP level (intensities) and how much variation exists.

We have included histograms for Brp::rGFP intensity and cluster volumes to Figure 3 in the revised manuscript.

(2) Line 52: "endogenous synapses" is a confusing term; it's probably meant that the protein levels within the synapse are endogenous and not overexpressed.

We apologize for the confusion and have revised the term to “endogenous synaptic proteins.”

(3) It is not clear from the Materials and Methods section, whether and where deconvolved or not-deconvolved images were used for the quantification pipeline. Please comment on this.

We have now revised the Method section to clarify how deconvolved or not-deconvolved images were differently used in the pipeline.

| (4) Line 664 (C) not bold.

We have corrected the error.

| (5) 725 "Files" should be Flies.

We have corrected the error.

| (6) 727 two times "first".

We have corrected the error.

| (7) Figure 7. All (A) etc., not bold - there should be consistent annotation.

We want to thank the reviewer for the detailed proof and have corrected all the errors spotted.

Reviewer #3 (Recommendations for the authors):

(1) Has there been an expression of the construct in a non-neuronal cell? Astrocyte-like cell? Any glia? As some sort of control for background and activity?

As the reviewer suggested, we verified the neuronal expression specificity of Brp::rGFP. Using *R86E01-GAL4* and *Amon-GAL4*, we compared Brp::rGFP in astrocyte-like glia and neuropeptide-releasing neurons. We found no Brp::rGFP puncta in the neuropils in astrocyte-like glia compared to neurons, suggesting Brp::rGFP is specific to neurons. We have included this new dataset to the revised manuscript (Figure 1 - figure supplement 2).

(2) Similarly, expression of the construct co-expressed with a channelrhodopsin, and induction of a 'learning'-like regime of activity, similarly in a control type of experiment, expression of an inwardly rectifying channel (e.g. Kir2.1) to show that increases in size of the BRP puncta are truly activity dependent? The NMJ may be an optimal neuron to use to see the 'donut' structures of the AZs and their increase with activity. Also, are these truly AZs we are seeing here? Perhaps try co-expressing cacophony-dsRed? If the GFP Puncta are active zones, then they should be surrounded by cacophony.

We would like to clarify that we did not find Brp::rGFP size increase upon learning. Instead, we demonstrated that associative training transiently remodelled sub-compartment-sized AZ "hot spots" in Kenyon cells, indicated by the correlation of local intensity and AZ density (Figure 6-7).

To demonstrate split-GFP tagging does not affect activity-dependent plasticity associated with Brp, we measured Brp remodeling in photoreceptors induced by constant light exposure (LL; Sugie et al., 2015 Neuron). Consistent with the previous study, we found that LL decreased the numbers of Brp::rGFP clusters in R8 terminals in the medulla, as compared to constant dark condition (DD). This result validates the synaptic plasticity involving dynamic Brp rearrangement in the photoreceptors (Figure 1F).

As the reviewer suggested, we performed the STED microscopy for the larval motor neuron and confirmed the donut-shape arrangement of Brp::rGFP (Wu, Eno et al., PLOS Biol 2025).

Also following the reviewer's suggestion, we double-labelled Brp::rGFP and Cac::tdTomato (Cacophony, the alpha subunit of the voltage-gated calcium channels). We found that 97% Brp::rGFP clusters showed co-localization with Cac::tdTomato in PAM-γ5 dopamine neurons terminals (Figure 1E), suggesting most Brp::rGFP clusters represent functional AZs.

| (3) *In the introduction: Intro, a sentence about BRP - central organiser of the active zone, so a key regulator of activity.*

We have included a few more sentences about the role Brp in the active zones to the revised manuscript.

| (4) *Figure 1 E, line 650 'cite the resource here'.*

We thank the reviewer for pointing out the error and we have corrected it.

| (5) *Many readers may not be MB aficionados, and to make the data more accessible, perhaps use a cartoon of an MB with the cell bodies of the neurons around the MB expressing the constructs highlighted so that the reader can have a wider idea of the anatomy in relation to the MB.*

We appreciate these comments and have appended cartoons of the MB to figures to help readers understand the anatomy.

<https://doi.org/10.7554/eLife.107663.2.sa0>